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REVIEW

Multi-method cooling strategies for photovoltaic systems: a comprehensive review of passive, active, and AI-optimized hybrid techniques

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Abstract

High operating temperatures significantly reduce photovoltaic (PV) system efficiency, lowering power output by up to 20%. This review examines passive, active, and hybrid PV cooling techniques addressing heat management challenges. Passive methods such as radiative cooling and phase change materials reduce PV temperature by up to 20 °C, improving electrical efficiency by 15.5%. Active cooling, including water and air systems, achieves larger temperature drops of up to 55 °C and electrical efficiency gains of up to 22.2%, albeit with higher operational costs. Hybrid approaches combine passive and active methods to optimize performance, balancing complexity, and energy savings. Artificial intelligence (AI)-based control techniques, including Reinforcement Learning, Long Short-Term Memory networks, and Genetic Algorithms, enable real-time optimization by adjusting cooling parameters based on environmental data. These AI methods, validated through experimental and simulation studies, have realized up to 6.7% energy savings and improved system durability by mitigating thermal stress. AI further supports predictive maintenance and adaptive cooling strategy enhancement, advancing smart PV cooling solutions. Despite these advances, challenges related to cost, scalability, and environmental impact persist. This review compares the performance and trade-offs of existing cooling technologies, identifies research gaps, and underscores the potential of integrated AI-driven hybrid systems. These insights contribute to developing next-generation, efficient, and sustainable PV cooling technologies.

1 Introduction

Photovoltaic (PV) systems are an important source of clean energy; nonetheless, their operation is impacted by high operating temperatures. Increased heat reduces the open-circuit voltage of PV modules and the overall efficiency of the modules, leading to real-world efficiency loss of up to 20% (Safar et al. 2020; Koohestani et al. 2023). Reliable cooling systems are thus essential for optimal performance and long-term durability of PV panels (Amr et al. 2024; Sultan et al. 2021).

Different cooling techniques have been researched, and all of them belong to the broad categories of passive, active, and hybrid cooling technologies. Passive cooling is through natural convection and radiation with no input of external energy, and methods like heat pipes, phase change material (PCM), and radiative cooling can cool PV by as much as 11 °C, with an improvement in electrical efficiency by as much as 44.12% under controlled PCM integration scenarios (Chandrasekar 2023; Kumar et al. 2024a). Active cooling, in contrast, depends upon external sources of energy such as forced convection and liquid cooling to augment heat transfer. Such technologies have the potential for up to 55 °C cooling reduction and up to 22.2% increase electrical efficiency gains of 22.2% from active water-spray cooling under standard testing conditions (Amr et al. 2024; Kumar et al. 2024a). Hybrid cooling technologies incorporate both passive and active methods, with the blend of heat pipes with forced convection or PCM with water cooling to serve the aim of energy consumption and efficiency balancing (Koohestani et al.

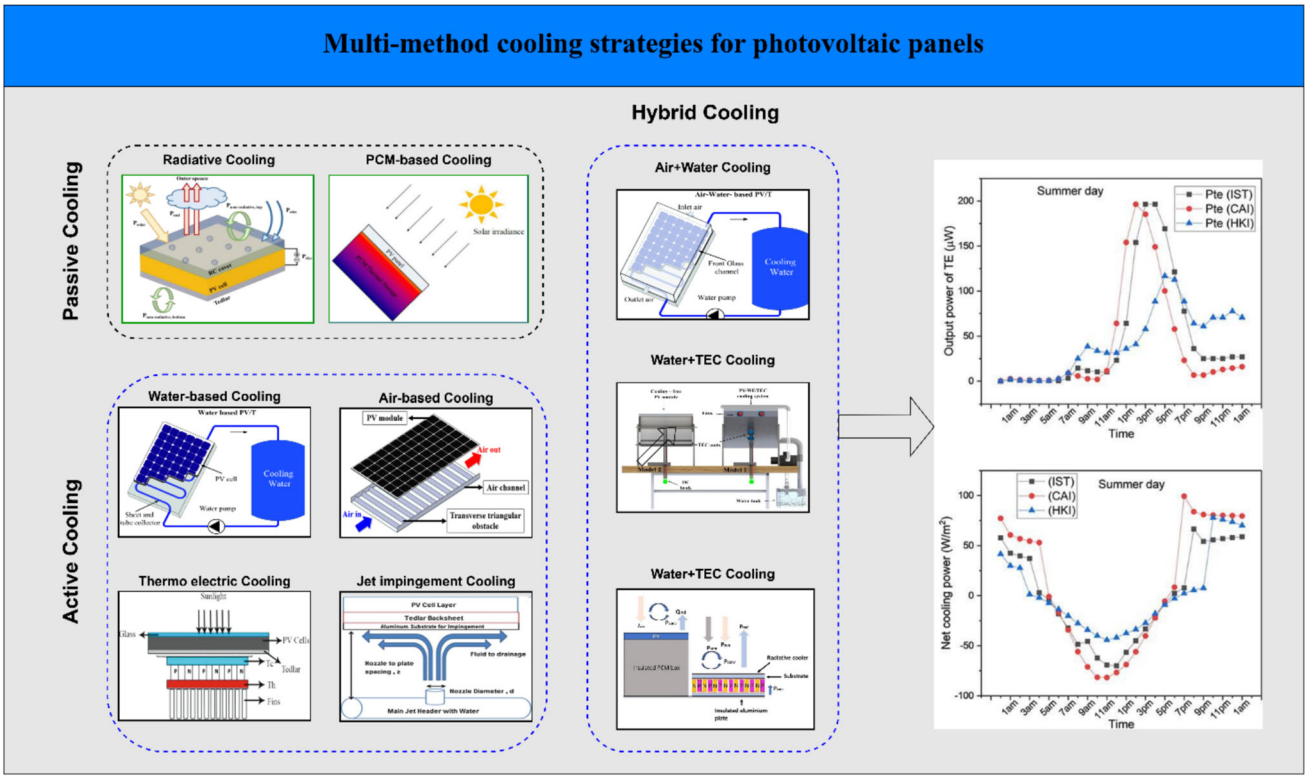
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Graphical abstract



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2023; Chandrasekar 2023). Due to the constraints of single-method systems, multi-method use is usually required to counteract different climatic conditions and working requirements (Kaneesamkandi and Rehman 2023; Wang et al. 2017). Hybrid cooling methods attain the best performance by leveraging the advantages of passive and active approaches, thus developing flexible and robust PV systems with the capacity to sustain performance under different environmental conditions (Zhuang et al. 2019; Mazón-Hernández et al. 2010).

Several review studies have analyzed photovoltaic (PV) cooling technologies, often focusing on a single category, such as passive cooling, AI-driven cooling, or water-based cooling. Studies on passive cooling emphasize its energy efficiency and sustainability. These methods include phase change materials (PCM), radiative cooling, and natural convection, all of which help in reducing PV temperatures (Akrouch et al. 2024; Harmailil et al. 2024). While these techniques reduce temperature without external power input, they often suffer from thermal conductivity issues and limited adaptability under fluctuating climatic conditions. In contrast, AI-enabled cooling methods, which are being extensively used in HVAC and industrial cooling, have shown potential in optimizing the cooling efficiency in real time using machine learning algorithms and predictive control methods (Zhou et al. 2023; Lu et al. 2025). Yet, existing reviews do not discuss the incorporation of AI into hybrid cooling methods, which can enhance energy saving and system flexibility for PV applications. Likewise, water-based cooling techniques, like water-spray and liquid immersion cooling, have been well reported, with up to 20% efficiency improvements but with water availability, maintenance, and operating cost concerns (Dwivedi et al. 2020; Pambudi et al. 2022).

Despite these contributions, significant research gaps remain. First, most reviews examine cooling methods in isolation, without considering the synergistic benefits of hybrid systems that combine passive, active, and AI-based strategies (Nasef 2019; Hasanuzzaman et al. 2016). Second, there are insufficient economic analyses and therefore it is challenging to estimate the cost-effectiveness and scalability of large-scale deployment of such cooling technologies (Salameh et al. 2021). Third, AI-based cooling applications in practice are still in their infancy and further research in dynamic optimization and automation for PV cooling is required. Addressing these gaps through a comprehensive,

multi-faceted strategy will assist in creating scalable and sustainable cooling solutions for photovoltaic systems.

Numerous studies have evaluated PV cooling techniques, assessing efficiency gains, limitations, and challenges. Research shows that air cooling, water-based cooling, PCM, nanofluids, and AI-assisted systems enhance PV performance under various climates. However, effectiveness is subject to environmental conditions, cost, and complexity of systems. Table 1 presents main findings, classifying cooling techniques according to efficiency enhancements, technical issues, and research gaps. Performance differs, ranging from water-spray cooling with 3.9% efficiency enhancement to hybrid and AI-based techniques offering up to 97.6%. Key limitations include the high cost, scalability problems, and environmental limits, calling for hybrid and AI-based long-term PV optimization approaches.

This review presents a comprehensive assessment of PV cooling techniques, addressing key gaps identified in previous studies. The main contributions of this study are:

- Comprehensive review of cooling methods: Compared with the previous studies of individual methodologies, this article has a wider-ranging comparative survey of passive, active, hybrid, and AI-based cooling methodologies concerning their strengths, weaknesses, and performance for varied operating environments.
- Analysis of performance under different environmental conditions: A significant portion of this review is the comparison of cooling techniques based on different climatic conditions, presenting an analysis of their performance concerning temperature range, humidity, and geographical location.
- AI and hybrid cooling integration development: AI-driven optimization of hybrid cooling systems is a less explored topic in the current literature, and this article addresses it. It highlights how AI can enhance adaptability, energy efficiency, and automated control for PV cooling applications.
- Economic and sustainability considerations: While most studies emphasize technical efficiency, this review also includes cost-effectiveness and environmental considerations in terms of highlighting the scalability and sustainability of coolers for widespread PV deployment.

By filling these key gaps, the present review moves forward the development of innovative photovoltaic cooling systems that integrate technical efficiency, economic viability, and automatic intelligence for improved performance and sustainability.

2 Classification of photovoltaic cooling techniques

Efficient cooling of photovoltaic (PV) panels is essential to maintain their performance and longevity. Cooling methods can be broadly classified into passive, active, and hybrid methods, each with its advantages and disadvantages, as depicted in Fig. 1.

2.1 Passive cooling techniques

Passive cooling methods offer a good means of heat reduction without actively consuming energy, and this can be useful in energy-efficient structures, electrical devices, and solar photovoltaic (PV) panels.

2.1.1 Radiative cooling

Passive cooling methods are preferred to prevent heat buildup without active energy, with applications in energy-efficient buildings, electronics, and solar photovoltaic (PV) systems. One promising passive method is radiative cooling, which has the potential to enable objects to emit heat to outer space passively (Harmailil et al. 2024; Kazem et al. 2023). Radiative cooling operates by emitting thermal radiation in the mid-infrared (MIR) range (8–13 μm), also known as the atmospheric transparency window. This allows heat to be dissipated directly into space without energy input (Kazem et al. 2023; Alao et al. 2025). One of the primary mechanisms of radiative cooling is the equilibrium of solar radiation received, atmospheric radiation, and emitted thermal radiation. Figure 2 shows the conceptual model of the process, with how the heat is both absorbed and dispersed by radiative and convective mechanisms.

The efficiency of radiative cooling is controlled by the Stefan–Boltzmann law, where the power emitted by a surface is:

$$P_{rad} = \epsilon \sigma {T_{sc}}^4 \tag{1}$$

where P_{rad} is the radiated power per unit area, ϵ is the surface emissivity, $\sigma = 5.67 \times 10^{-8}\ \mathrm{Wm^{-2}\,K^{-4}}$ is the Stefan–Boltzmann constant, and T_{sc} is the emitting surface temperature (Fang et al. 2024). Radiative cooling efficiency is reliant on the capacity of a material to selectively emit in the MIR region of radiation and suppress solar range (0.3–2.5 μm) of absorption (Ahmed et al. 2021).

Table 1 Summary of previous research on photovoltaic cooling techniques and determined research

Study (Year)	Cooling method	Methodology	Key findings	Efficiency improvement (%)	Limitations/gaps identified	Refs.
2024	Water-spray cooling	Experimental	Front surface water spray enhances efficiency via thermal regulation and cleaning effects	3.9%	Limited to specific conditions; high water use in arid regions	Basem et al. (2024)
2022	Dual-cooling system (TEC + WFCS)	Experimental	TEC and water flow cooling lower module temperature to 21 °C, boosting efficiency	11.23%	System complexity and cost; large-scale feasibility unknown	Eid et al. (2022)
2017	Phase change materials (PCM)	Literature review	PCM integration enhances electrical efficiency up to 5%	Up to 5%	Poor conductivity and high cost; lack of reliable design standards	Chandel and Agarwal (2017)
2021	Jet impingement cooling	Simulation and experimental	Jet impingement in PV-wind hybrids yields higher cooling efficiency than conventional systems	–	Optimization needed for climate-specific applications	Siah Chehreh Ghadikolaei (2021)
2023	Radiative cooling + heat sink	Experimental	Desert-specific system combining radiative cooling and heat sinks reduces temperature and raises efficiency	11.11%	Needs validation across diverse climates	Kumar et al. (2023a)
2024	PCM + thermoelectric cooling (TEC)	Experimental	Combined PCM and TEC system reduce temperature significantly, improving efficiency	9.69%	Fixed PCM encapsulation limits scalability	Khan et al. (2024)
2018	Water-based cooling	Literature review	Various water-based cooling approaches can increase PV efficiency by 10–20%	10–20%	Economic and environmental impacts insufficiently analyzed	Nižetić et al. (2018)
2024	Various cooling methods (water, air, PCM, nanofluids)	Literature review	Cooling techniques improve efficiency from 0.28 to 97.6%, with temperature drops up to 39.9 °C	0.28–97.6%	Need for comprehensive performance, economic, and environmental evaluation	Sohail et al. (2024)

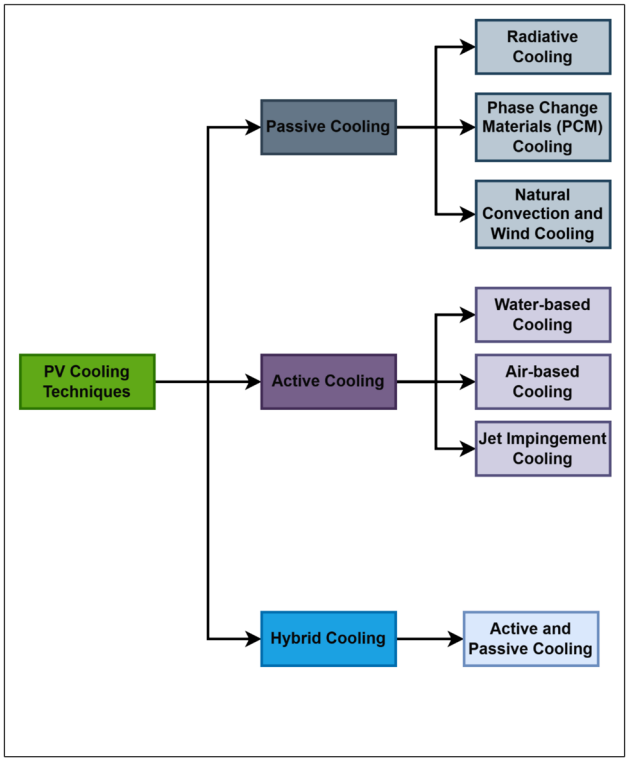


Fig. 1 General classification of PV cooling techniques

Another basic equation of radiative cooling considers the net surface cooling power, which is the equilibrium of radiation emission, atmospheric absorption, solar absorption, and convective heat transfer:

$$P_{net} = P_{rad} - P_{atm} - P_{sun} - P_{conv} \tag{2}$$

where P_{atm} is the atmospheric radiation absorbed by the surface, P_{sun} is the solar radiation absorbed, and P_{conv} represents convective heat transfer. Additionally, the emissivity of the sky can be estimated using:

$$\mathcal{E}_{sky} = 0.711 + 0.56\left(\frac{T_d}{100}\right) + 0.711\left(\frac{T_d}{100}\right)^2 \tag{3}$$

where T_d is the dewpoint temperature, which is calculated based on ambient conditions:

$$T_d = \frac{237.7r}{17.27.r} \tag{4}$$

$$r = \frac{17.27(T_{amb} - 273)}{23.77 + T_{amb} - 273} + In(RH). \tag{5}$$

These equations help estimate the radiative cooling potential under different atmospheric conditions (Akerboom et al. 2022).

Environmental factors such as humidity, dust accumulation, and cloud cover limit the performance of radiative cooling by reducing the escape of infrared radiation (Fang et al. 2024; Bamisile et al. 2025). Figure 3 illustrates temperature reduction comparisons between radiatively cooled and conventional PV modules. In Fig. 3a,

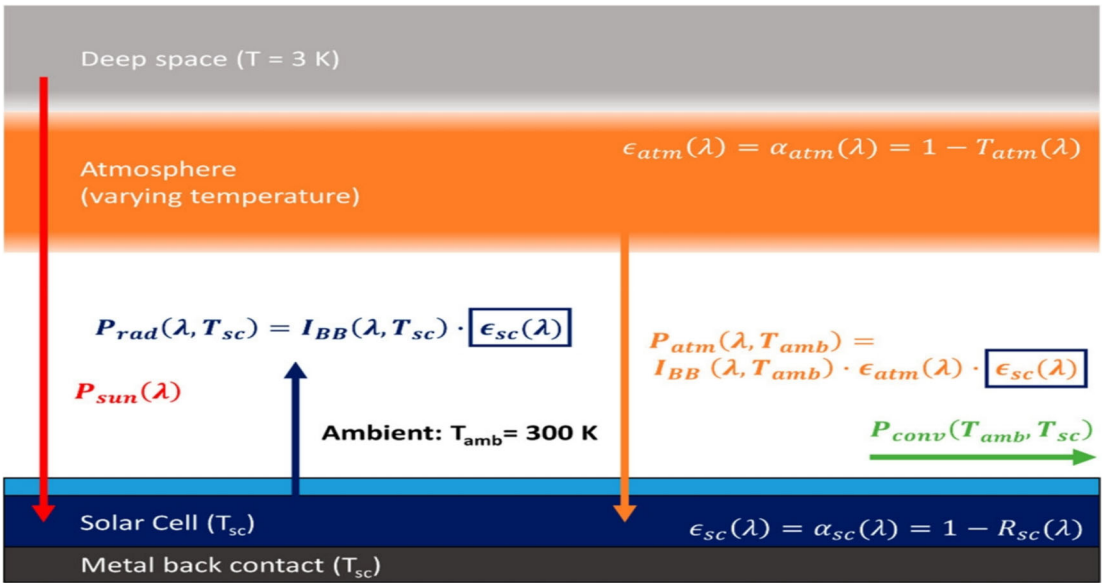


Fig. 2 Schematic drawing that shows the equality of in- and out-going power flows which determine the solar cell's temperature of equilibrium, T_sc. The solar cell is in thermal equilibrium when absorbed solar power P_sun and atmospheric thermal radiation P_atm are balanced by emitted thermal radiation P_rad from the solar cell and by power P_conv lost by convection and conduction (Source: Adapted from Akerboom et al. 2022)

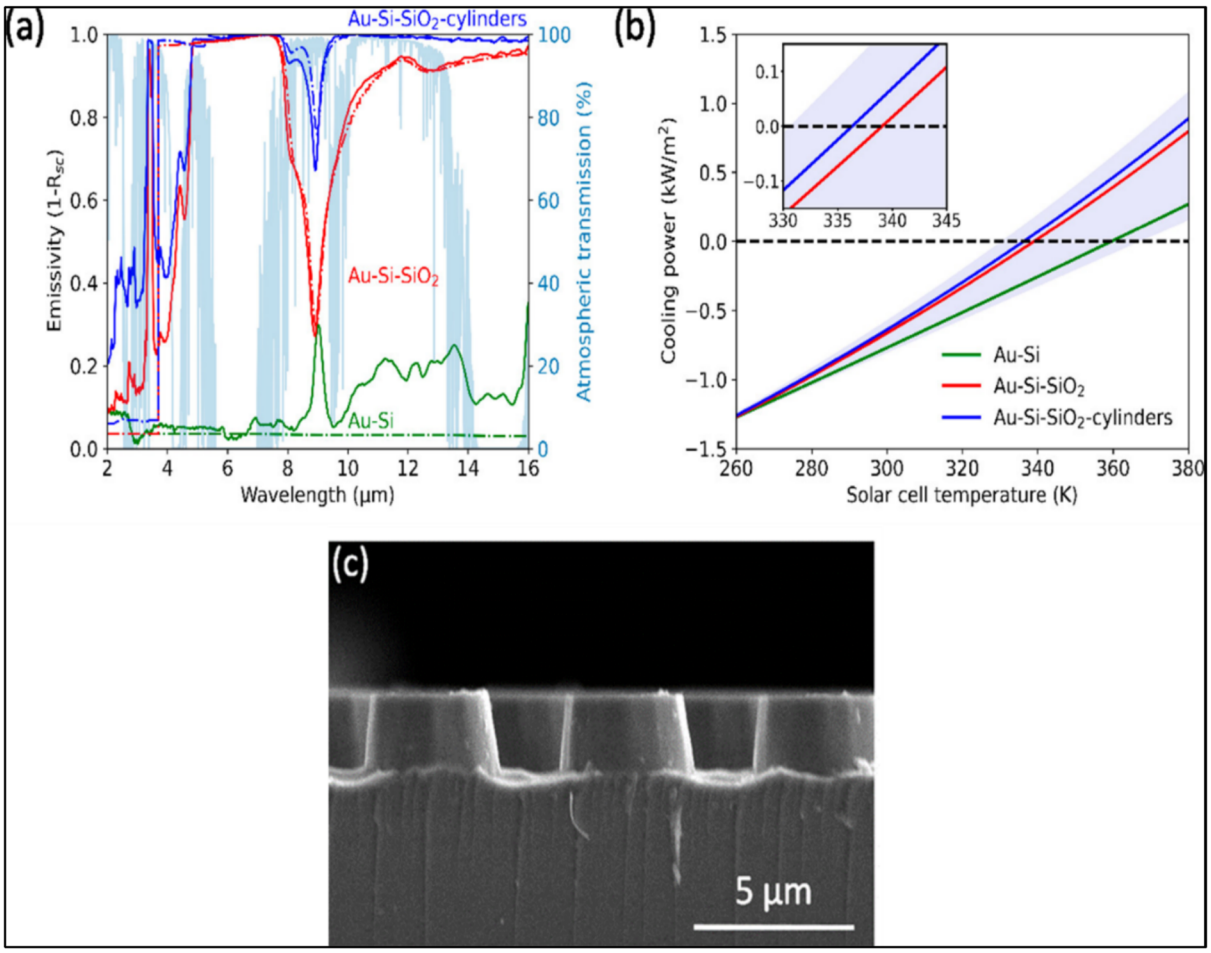


Fig. 3 Comparison of the temperature decrease by reference PV modules and radiative cooling PV modules. **a** Emissivity spectra for various PV geometries, indicating the enhanced radiative cooling function of Au–Si–SiO_2-cylinders, **b** Cooling power and solar cell temperature, which indicates the enhanced cooling effect of micro-structured surfaces. **c** SEM image of the cross-section morphology of PV modules improved using microcylinder, which is consistent with enhanced infrared emissivity and heat dissipation (Source: Adapted from Akerboom et al. 2022)

the spectra of emissivity indicate that Au–Si–SiO_2-cylinders are beneficial for heat dissipation. Figure 3b illustrates structured surfaces enhance cooling power with increasing solar cell temperature. Figure 3c illustrates an SEM micrograph of microcylinder-enhanced PV modules with enhanced emissivity and thermal management (Akerboom et al. 2022). Radiative-cooled PV modules show a 0.043% annual improvement in stability due to sustained temperature reduction, which mitigates long-term thermal stress (Sun et al. 2017). Additionally, micro-structured coatings and photonic materials preserve high mid-infrared emissivity despite environmental degradation from aerosols and weathering, supporting system durability over time (Akerboom et al. 2022).

Radiative-cooled PV modules exhibit improved thermal regulation, reducing operating temperature and slowing material degradation. For instance, a V-shaped PV structure decreased temperature by 10.6 °C and increased power output by 16.8% (Fang et al. 2024), while a photonic silica microcylinder lowered temperatures by 3 °C and improved efficiency by 1.35% (Akerboom et al. 2022). Cooling power per unit area is a key metric for radiative cooling performance. Figure 4 presents spectral and cooling efficiency differences among materials and coatings. Figure 4a compares absorptivity and emissivity spectra. Figure 4b shows how cooling power varies with temperature. Figure 4c highlights spectral irradiance from sky and ground heat sinks (Fang et al. 2024). A multilayer soda-lime glass structure demonstrated cooling power of 75 W/kg—3.7 times more efficient than conventional copper coolers (Zhu et al. 2014).

Various materials enhance radiative cooling in PV modules, including ceramic-based, perovskite, and hydrogel-based coatings. Ceramic coatings such as Al_2O_3, Si_3N_4, and TiO_2 improve emissivity, reducing temperatures by 5–10 °C

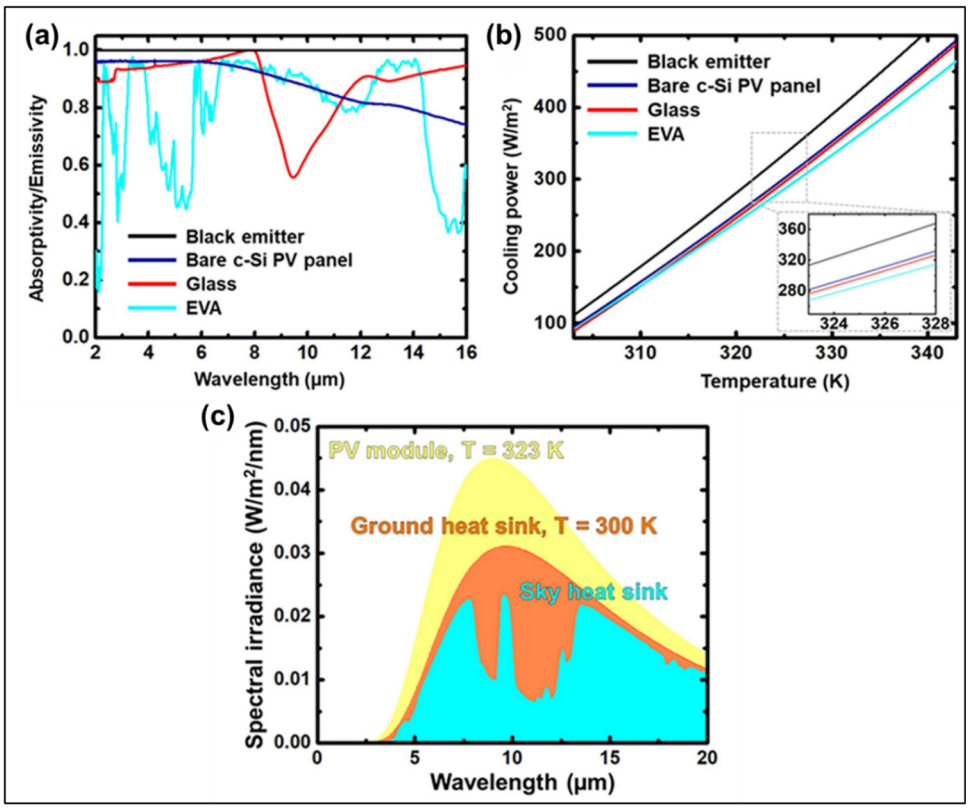


Fig. 4 Comparison of radiative cooling efficiency of various materials and coatings. **a** Absorptivity and emissivity spectra of various PV module configurations with spectral characteristics of black emitters, uncoated silicon PV panels, glass, and EVA layers. **b** Cooling power versus temperature, demonstrating relative efficiency of various materials for thermal management. **c** Spectral irradiance of various heat sinks, showing the ability of sky and ground heat sink radiative cooling. (Source: Adapted from Fang et al. 2024)

and boosting efficiency by 10% (Al-Dharob et al. 2024), as illustrated in Fig. 5. However, TiO_2 has the drawback of UV-induced aging (Li et al. 2024). ZrO_2 and $KZr_2(PO_4)_3$ have high emissivity and low absorption and are, therefore, used at high temperatures (Yao et al. 2015), and Al_2O_3 is used in space technology for thermal control (Cao et al. 2025; Ghanim et al. 2025).

Radiative cooling offers a promising passive approach for managing PV module temperatures, enhancing energy output and system longevity by minimizing thermal cycling stress. Ceramic and perovskite-based photonic coatings have demonstrated significant potential by combining high mid-infrared emissivity with solar-range reflectivity. However, the performance of radiative cooling can be affected by environmental factors such as humidity, aerosol deposition, and dust accumulation, which reduce the escape of infrared radiation. Maintaining high mid-infrared emissivity while minimizing solar absorption under real-world environmental conditions remains a significant challenge. To overcome this, advanced material designs featuring self-cleaning surfaces, durable coatings, and optimized angular emissivity are under development to sustain long-term cooling performance. Alternative passive cooling techniques, such as phase change materials (PCMs) and hydrogel-based evaporative systems—which rely on latent heat transfer mechanisms—are discussed in subsequent sections to highlight their distinct operational principles.

2.1.2 Phase change materials (PCM) cooling

Phase change materials (PCMs) are suited to control the temperature of photovoltaic (PV) panels through heat absorption and release over the phase transition. PCMs experience a solid–liquid phase transition, absorbing heat and liquefying by day and freezing overnight and emitting stored heat energy at almost constant temperatures (Socaciu 2012; Mousavi Baygi and Sadrameli 2018). This passive level of thermal control stabilizes the temperature of panels, averting overheating which lowers PV performance (Sellami et al. 2017). While higher amounts of energy are in liquid–gas transitions, they are less commonly used in PV cooling because they require higher temperature and are cumbersome to handle the gas phase (Kazem et al. 2023).

PCMs mainly function by absorbing and releasing heat. The material absorbs the surplus heat from the PV panel in the daytime and preserves the heat as latent heat when it changes phase from solid to liquid (Mousavi Baygi and Sadrameli 2018). The absorbed heat is released at night or in cold climates when the PCM changes back to solid, sustaining a constant temperature (Sellami et al. 2017). This thermal energy exchange follows the equation:

$$Q = m.L \tag{6}$$

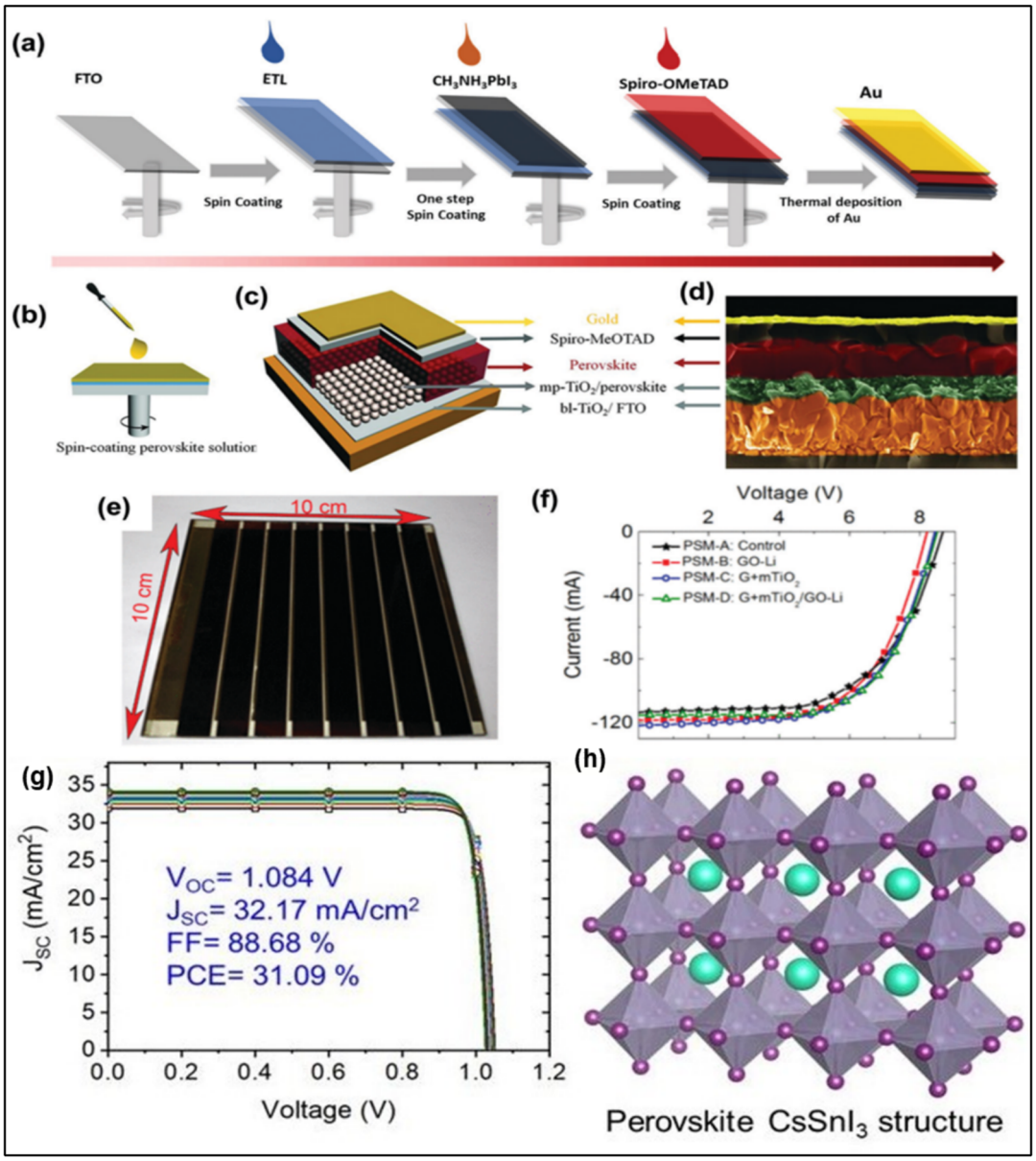


Fig. 5 Detailed illustrations and performance analysis of perovskite coatings used in radiative cooling applications: **a** Spin-coating of the electron transport layer (ETL) followed by the deposition of the $CH_3NH_3PbI_3$ perovskite layer. **b** Addition of hole transport layer (HTL) and gold electrode, completing the multilayered PV structure. **c** Cross-sectional schematic emphasizing TiO_2 and fluorine-doped tin oxide (FTO) integration, enhancing mid-IR thermal emission. **d** SEM micrograph showing the internal nanostructure that facilitates efficient radiative heat dissipation. **e** Fabrication of a scalable large-area perovskite solar module optimized for radiative cooling applications. **f** I–V characteristics comparing unmodified and optimized modules, demonstrating improved electrical performance. **g** Lead-free perovskite solar cell configuration, highlighting environmentally safe design with high optical transparency. **h** Molecular structure of $CsSnI_3$ perovskite, illustrating its tunable bandgap for enhanced photonic and cooling functionality. Source: Adapted from (Fatima et al. 2023; Shalan 2020)

where: Q is the heat absorbed or released (Joules), m is the PCM mass (kg), L is the latent heat of fusion (J/kg) (Tawalbeh et al. 2023).

A variety of PCMs have been used for PV cooling, including paraffin wax, Glauber's salt, sodium acetate trihydrate, and fatty acids such as palmitic and stearic acid (Karthick et al. 2023; Prabhu et al. 2024; Kenisarin and Mahkamov 2007; Dutt Sharma et al. 2004). Paraffin PCMs are likely to be utilized for high latent heat capacity with low thermal conductivity, which may be improved by introducing nanoparticles like TiO_2–Ag composites (Sen and Celiktas 2024). Salt hydrates like $CaCl_2{\cdot}6H_2O$ have the potential

based on their enormous heat storage with appropriate melting point for the application of PV cooling (Díaz et al. 2022). Additionally, composite PCMs, such as Hexadecanol-Palmitic Acid/Expanded Graphite (HA-PA/EG), demonstrate improved thermal conductivity and enhanced stability (Xu et al. 2024).

The effectiveness of PCM cooling in lowering PV panel temperatures and improving electrical efficiency is well-documented. PCMs can reduce PV panel temperatures by 3–26.6 °C, depending on the PCM type and configuration used (Prabhu et al. 2024; Sen and Celiktas 2024; Díaz et al. 2022). Electrical efficiency improvements range typically from 1 to 30%, with some studies reporting electrical efficiency improvement up to 56% under optimized PCM configuration and irradiance level (Karthick et al. 2023; Prabhu et al. 2024). For instance, sodium acetate trihydrate and palmitic acid demonstrated a 22.84% temperature decrease leading to a 24.95% electrical efficiency increase (Sen and Celiktas 2024). Similarly, polyethylene glycol 1000 (PEG1000) showed a temperature reduction of 15 °C and an 8% electrical efficiency enhancement (Mousavi Baygi and Sadrameli 2018).

A further experimental study by Xu et al. (2023) examined PV parameters under PCM cooling, including temperature change, short-circuit current (Isc), open-circuit voltage (Voc), fill factor (FF), and maximum power output (Pmax). Figure 6a shows a temperature decrease of 36.51 °C, confirming effective thermal regulation. Figure 6b displays higher Voc and Isc, minimizing efficiency losses from overheating. Figure 6c indicates stable or improved FF, while Fig. 6d shows an increase in Pmax by 1.35 W, corresponding to a 1.63% overall electrical efficiency gain.

Compared to conventional air- and water-cooling systems, PCM cooling provides a reliable, passive thermal regulation method that requires no external energy input. While water-cooling can improve PV electrical efficiency by 1.6–3.8%, its practical use is limited in arid regions due to water availability (Bhakre et al. 2023). PCM cooling achieves more substantial temperature stabilization and electrical efficiency improvements over long-term operation (Sahu et al. 2021).

Despite benefits, PCM cooling faces challenges such as low thermal conductivity, leakage, and phase segregation. Advances including nanoparticle additives (CuO, Al_2O_3, ZnO), metallic fins, porous foams, and multi-layer PCM composites have significantly improved heat transfer and system stability (Seto et al. 2024; Variji et al. 2022; Sheikh et al. 2024a). Table 2 summarizes key experimental and computational studies on PCM cooling for PV panels, demonstrating temperature reductions from approximately 5 to 37 °C and electrical efficiency gains typically ranging from 1.3 up to 17%, depending on PCM type, system configuration, and environmental conditions.

PCM-based cooling has emerged as a promising passive thermal regulation technique for photovoltaic (PV) panels, effectively reducing operating temperatures and enhancing electrical efficiency by mitigating losses due to overheating. Various PCMs—including paraffin wax, salt hydrates, and composites—have demonstrated temperature reductions typically between 5 and 37 °C, with electrical efficiency gains ranging from about 1 to 17%, depending on material properties and system design. Despite these advantages, PCM cooling faces challenges such as low thermal conductivity, phase segregation, and leakage, which can affect long-term reliability. Advances in material science—including nanoparticle additives, metallic fins, and hybrid PCM configurations—have significantly improved thermal performance and system stability. Ongoing research focuses on integrating PCMs with complementary cooling technologies such as thermoelectric coolers and liquid cooling, as well as developing novel heat discharge mechanisms. These efforts aim to optimize PCM formulations and enable low-cost, scalable thermal management solutions suitable for widespread solar industry adoption, particularly in regions with high solar irradiance.

Phase change materials (PCMs) encounter challenges such as phase segregation, leakage, and diminished latent heat capacity due to repeated thermal cycling. These degradation effects can reduce long-term cooling efficiency and necessitate consideration of replacement or maintenance cycles in practical applications. Although comprehensive lifecycle assessments (LCAs) are currently limited, incorporating durability and lifecycle aspects is essential for a realistic evaluation of PCM-based cooling technologies. Addressing these durability challenges is critical for the development of sustainable, long-lasting PV cooling solutions.

2.1.3 Natural convection and wind cooling

Natural convection and wind cooling are passive methods that use airflow to control the temperature of photovoltaic (PV) panels. The methods improve heat dissipation by inducing air flow over and around the PV surface, minimizing excessive heating and electrical efficiency (Dwivedi et al. 2020; Akrouch et al. 2023). The efficiency of these cooling techniques relies on a variety of factors such as wind speed, panel tilt, and ventilation design. Natural convection occurs because of movement of air due to temperature gradients, inducing flow caused by buoyancy on the PV surface (Kandeal et al. 2024). Several researchers have experimented with air gaps, sloping inlet channels, perforated frames, and extended ducts with the expectation of enhancing heat dissipation to achieve higher efficiency. It has been found through research that the inclusion of a 120 mm air gap in an insulated channel improved daily energy output by 1.7% and efficiency

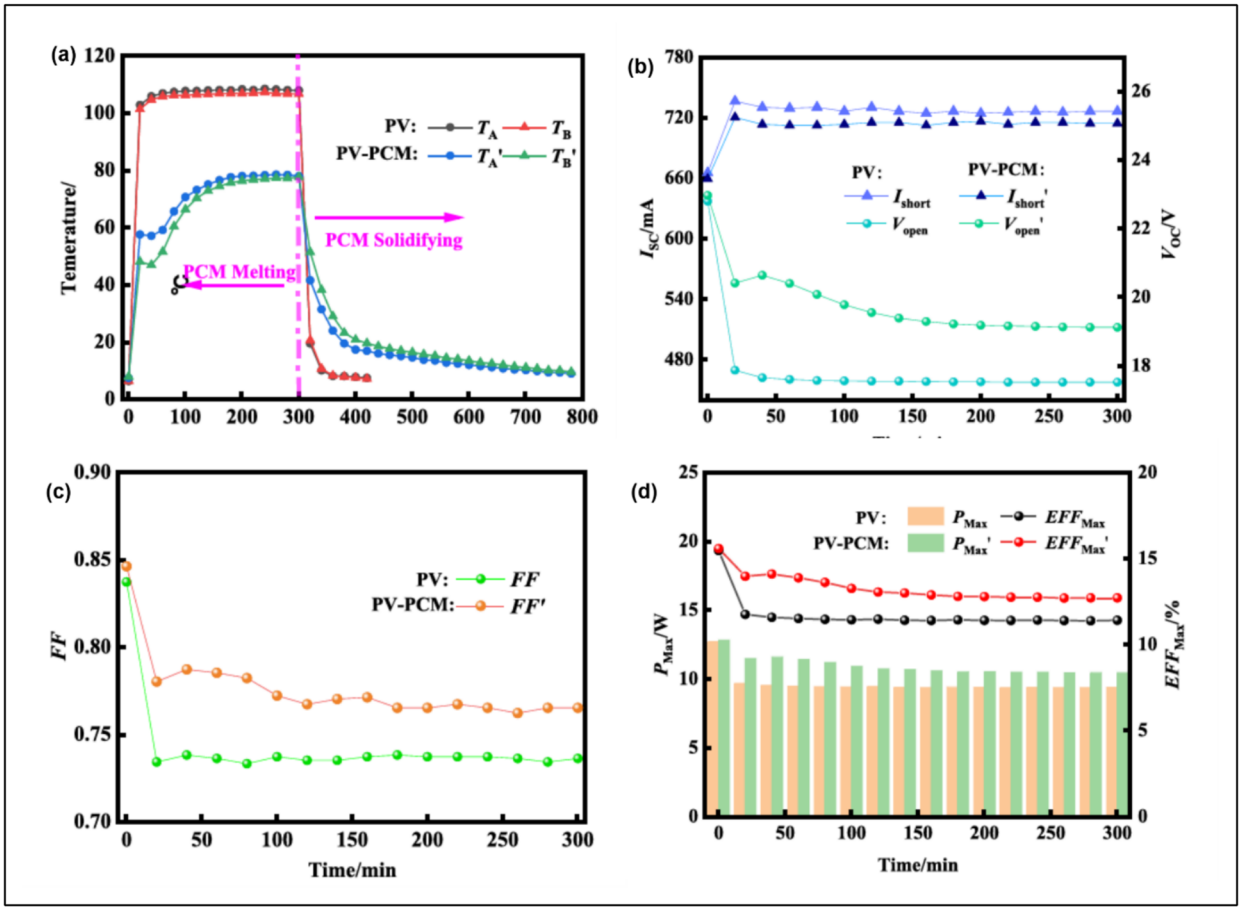


Fig. 6 Performance Analysis of PCM Cooling on PV Panels. **a** Temperature variation of PV panels with and without PCM cooling, **b** Changes in short-circuit current (Isc) and open-circuit voltage (Voc), **c** Fill Factor (FF) variation, **d** Maximum power output (Pmax) comparison (Xu et al. 2023)

by 1.2% (Almuwailhi and Zeitoun 2023). Additionally, the use of topology-optimized copper sheets resulted in a 2.7 °C temperature reduction and a 1.30% increase in power output, while perforated heat sinks enhanced PV efficiency by 5.28–5.92% (Hudișteanu et al. 2025).

Wind-induced cooling utilizes external airflow to augment convective heat release. Some of the modifications that have been introduced on rooftop PV systems include vortex generators and curved eaves, resulting in an optimum temperature drop of 5.89 °C and better power conversion efficiency (PCE) (Wang et al. 2023a). Wind tunnel testing reveals that elevated wind speeds (e.g., 6 m/s) can decrease PV temperatures by as much as 12.7 °C (Zhou et al. 2024). Table 3 summarizes recent research exploring natural convection and wind-induced cooling methods for PV panels, covering cooling methods, methodology, system design, and main performance enhancements like temperature drop, power output, and efficiency increase.

Natural convection and wind-based cooling techniques are energy-saving and low-cost techniques to enhance PV performance. Natural convention depends on design changes such as air gaps, perforated frames, and topology-optimized materials to enhance heat transfer while wind-based cooling makes use of external flow to increase convective heat loss. Studies indicate that temperature reductions ranging from 2.7 to 12.7 °C can be achieved, leading to efficiency improvements of up to 5.92%. However, the effectiveness of such methods relies on climatic conditions, and areas having low wind speeds or stagnant air benefit little or not at all from passive cooling systems. In upcoming studies, hybrid systems combining natural convection with other active and passive cooling systems must be developed to improve PV panel efficiency across various conditions.

Many passive cooling techniques demonstrate promising performance at the laboratory scale; however, scaling these methods to utility-scale photovoltaic (PV) installations presents significant challenges that remain inadequately

Table 2 Recent studies on cooling photovoltaic panels using composite phase change materials (PCMs)

Refs.	Year	PCM used	Methodology	Configuration	Key results
Sen and Celiktas (2024)	2024	Paraffin PCM with metal oxide nanoparticles	Experimental	PCM slabs on PV backside	Temperature reduced by approximately 15 °C; energy yield improved by 1.3–2.5%
Khan et al. (2024)	2024	PCM combined with Thermoelectric Cooler (TEC)	Experimental	Integrated PV/PCM/TEC system	Maximum temperature reduction of 9.28 °C; power output increased by 9.69%
Sharaf et al. (2022)	2022	Hybrid PCMs	Experimental	PV panels with different PCM thicknesses	Temperature reduction between 4.9 and 8.1 °C; electrical efficiency increased by 5.0–7.8%
Mohammadifar et al. (2025)	2024	Paraffin PCM with multi-walled carbon nanotubes (MWCNT)	Experimental	PV/T system with multi-block fined absorber	Energetic efficiency improved by 13.25%; estimated CO_2 reduction of 18.01 tons
Mohammadifar et al. (2025)	2025	PCM with flattened heat pipes (FHPs)	Experimental	PV panel with FHPs and graphite sheet	Temperature reduced by up to 37.0 °C; electrical efficiency increased by 17.3%
Maghrabie et al. (2023)	2023	Pure beeswax blends	Experimental	PCM housed behind PV panels	Temperature reduction of 19%; electrical power production increased by 13%
Elsabahy et al. (2022)	2022	Calcium chloride hexahydrate ($CaCl_2 \cdot 6H_2O$)	Simulation and experimental	PV-PCM system tested in Vicuña and Calama	Temperature reduced by up to 17.5 °C; electrical output improved by 4.5–5.8%

addressed. Key issues such as pressure losses, flow maldistribution, and increased thermal resistance can substantially reduce cooling efficiency when phase change materials (PCMs), thermal masses, or natural and forced convection systems are deployed over large areas. These challenges introduce additional operational complexity, requiring meticulous hydraulic design and precise flow management to ensure uniform cooling performance. To overcome these obstacles, comprehensive computational fluid dynamics (CFD) modeling combined with pilot-scale experimental validation is essential for understanding flow dynamics and optimizing system design. Addressing these scale-up factors is critical for enabling the practical, efficient, and reliable implementation of passive cooling technologies in large-scale solar power plants.

Despite promising results from numerous experimental studies, several limitations affect the generalizability and practical application of these findings. Many investigations utilize small sample sizes and short testing durations that may not capture long-term system stability or performance under variable environmental conditions. Furthermore, the absence of standardized testing protocols often hampers direct comparison and reproducibility across different research efforts. Additionally, the integration of cooling solutions with actual photovoltaic modules in real operational settings is frequently limited, raising questions about scalability and durability. These challenges underscore the need for comprehensive, large-scale, and long-term field evaluations to robustly validate the effectiveness, reliability, and economic feasibility of passive cooling methods before they can be widely implemented in commercial PV systems.

2.2 Active cooling techniques

2.2.1 Water-based cooling

Water cooling is one of the efficient techniques for the performance improvement of photovoltaic (PV) panels by eliminating excess heat through water flow or spraying. Cooling maintains the temperature at a constant level, preventing efficiency loss through overheating. Water cooling

Table 3 Recent studies on cooling photovoltaic panels using natural convection and wind-based cooling

Refs.	Year	Cooling approach	Methods	Configuration	Results
Almuwailhi and Zeitoun (2023)	2023	Natural convection	Insulated channel with varying air gaps (30, 60, 90, 120 mm)	PV panel with 120 mm air gap	Increased daily energy production by 1.7%, efficiency by 1.2%
Wang et al. (2023a)	2023	Wind-based cooling	Curved eaves and vortex generators (VGs)	Rooftop PV with curved eaves and VGs	Maximum temperature reduction of 5.89 °C, improved PCE
Lebbi et al. (2024)	2024	Natural convection	Sloped entrance channel (SEC) and aluminum flat plate (AFP)	Hybrid PV/T system with SEC and AFP	Improved heat transfer rate by up to 20%
Nazari and Eslami (2021)	2021	Natural convection	Perforated aluminum frame	PV module with 10 mm circular perforations	3.8 K temperature drop, 0.62% improvement in power generation
Ganesan et al. (2022)	2022	Natural convection	Black coated extended duct	PV panel with extended duct	Significant enhancement of current, voltage, and overall efficiency
Tripanagnostopoulos and Themelis (2010)	2010	Natural convection	Thin metal sheet (TMS) and metal fins (FIN)	PV panels with TMS or FIN in air duct	Improved electrical performance due to reduced temperature
Zhou et al. (2024)	2024	Natural convection	Topology-optimized copper sheet	PV wall with copper sheet	Temperature reduction of 2.7 °C, output power increment rate of 1.30%
Hudișteanu et al. (2025)	2025	Natural convection	Heat sinks with fins	PV panels with perforated and non-perforated fins	Efficiency improvements of 5.28–5.92%
Han et al. (2019)	2019	Natural convection	Passive air channel	PV façade with open and closed channels	Maximum surface temperature of 49.1 °C for open channel

has been shown in research studies to enhance PV efficiency by 10–20% (Dwivedi et al. 2020) and thus is a viable method of power enhancement. Excessive water usage and scaling/fouling of PV surfaces, however, continue to be the most significant issues that need to be resolved for long-term sustainability.

Several water-based cooling techniques have been developed, each varying in cooling efficiency and implementation complexity. One of the most applied methods is water circulation, in which water is continuously circulated through copper tubing or an insulated tank at the back of the PV panels. As illustrated in Fig. 7, the system integrates various components such as the water circulation loop, PVT collector, and storage tank to manage both electrical and thermal energy, with water-based cooling used to reduce the temperature of the PV panels. Researchers have found that the application of copper tubing with circulating water can decrease temperatures by 15.23% and improve electrical efficiency by 6.08% (Singh et al. 2020). A method, which employed a tank with an insulated closed water loop, showed good extended cooling and uniform temperature distribution (Sonia et al. 2018). Another common method that has been widely researched is water-spray cooling, where water is sprayed over the front side of PV panels. According to research, it can enhance efficiency by 3.9% and keep the temperature at 41 °C (Basem et al. 2024). Separate research also recorded a 15.3% rise in power and a decrease in temperature from 53 to 23 °C with the utilization of a single nozzle spray (Khan et al. 2023). Additionally, based on the orientation of the nozzle and ambient condition, this technique can lead to as much as 22.72% efficiency improvement (Navaei et al. 2024).

Moreover, water spray cooling not only provides thermal regulation but also contributes to panel cleanliness by removing dust and other particulate matter, resulting in a self-cleaning effect that maintains panel efficiency. Studies have demonstrated efficiency improvements of up to 22.72% attributed to this cleaning alongside cooling (Navaei et al. 2024). The utilization of air conditioning condensate water for cooling has also shown a 14% increase in power output due to combined cooling and cleaning effects (Shah

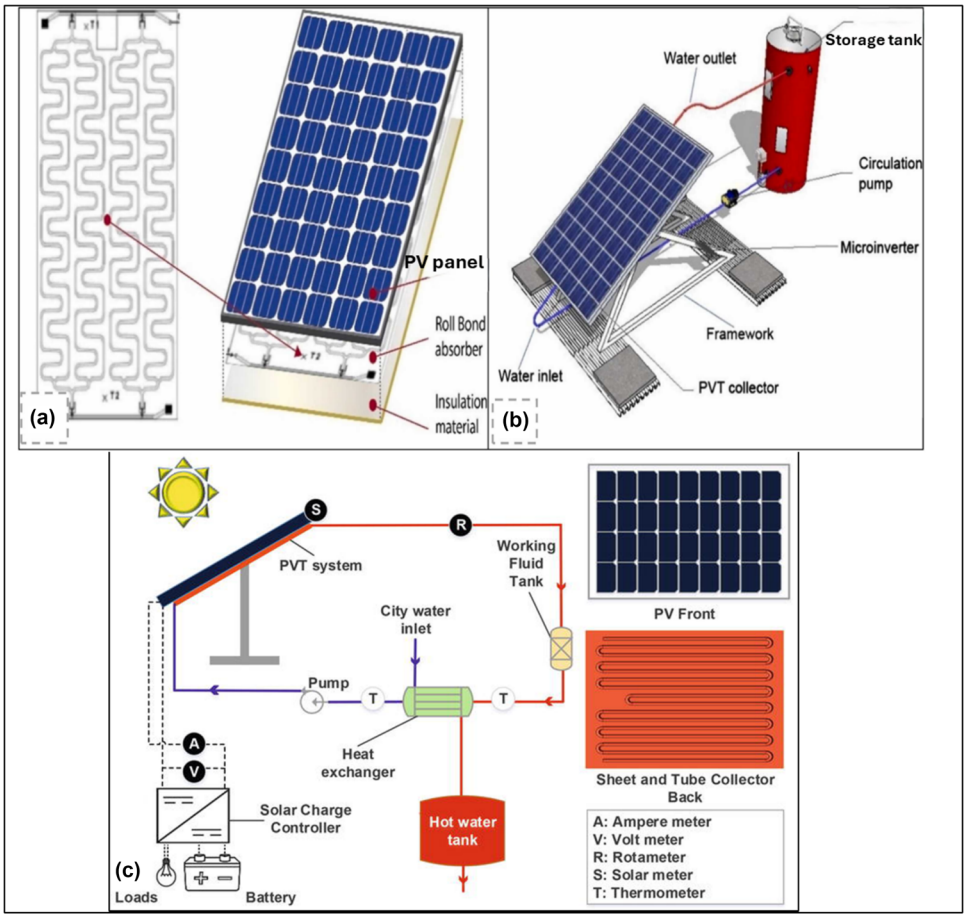


Fig. 7 Water-based cooling system for photovoltaic panels. **a** Water circulation system with copper tubing behind the PV panel for heat absorption. **b** Integrated system with PVT collector, circulation pump, and storage tank (Source: Adapted from (Abd-Elhady et al. 2025)). **c** System flow diagram showing the connection to the heat exchanger and hot water tank (Source: Adapted from Sardarabadi and Farzanehnia 2019)

et al. 2023). This interaction between water-based cooling and soiling management enhances energy yield and reduces manual cleaning requirements.

Water flow cooling is yet another passive method whereby a thin layer of water flows over the surface of the panel to carry away excess heat. The system, which can be automated through the utilization of a microcontroller, has been shown to enhance power output by 10.30% and electrical efficiency by 19.8% (Dewi et al. 2024). Additionally, tests have also proved that utilization of different water flow rates can cause power improvement of up to 11.99%, which reveals the cost-effectiveness of the method at low flow rates (Moraes et al. 2017). Passive cooling techniques through flowing water films over the PV surface also showed to improve efficiency by 15.28–17.75%, as a promising alternative to conventional active cooling systems (Tashtoush and Al-Oqool 2019). Nanofluid cooling is another technique, in which nanoparticles are dispersed in the cooling fluid to enhance the heat transfer. Studies have demonstrated that using water with Al_2O_3 nanofluid at various concentrations and flow rates achieved a 23.14% temperature decrease and a 13% power improvement at a 3 wt% concentration (Murtadha and Hussein 2022). Similarly, another study using a mixture of Al_2O_3 and TiO_2 nanofluids reported a 25% temperature reduction and a 10.3% increase in electrical efficiency (Dev et al. 2022), proving that nanofluid-based cooling has strong potential for improving PV performance.

Apart from active water cooling, a more sophisticated method, referred to as forced water cooling, has been investigated. This method, involving the integration of serpentine cooling blocks or in-wall embedded heat exchangers, greatly increases the heat dissipation rate and has been applied to raise PV output power by 14.29% (Sheikh et al. 2020). All these methods are shown in Table 4, which gives the recent studies investigating water-based cooling methods, approaches and system design, and performance enhancement.

In addition to boosting PV efficiency, water-based cooling is confronted with excessive water consumption in arid environments and efficiency degradation due to scaling and fouling. These aspects increase the cost of maintenance and large-scale application. Closed-loop systems, self-cleaning surfaces, and sophisticated filtration can reduce scaling impacts through optimizing water consumption. Hybrid nanofluids and PCMs can be incorporated to enhance thermal regulation while reducing water dependency. AI-optimized

Table 4 Recent studies on water-based cooling for photovoltaic panels

Refs.	Year	Cooling approach	Methodology	Configuration	Key findings
Singh et al. (2020)	2020	Water circulation	Experimental	Copper tubes attached to rear of PV panel	Reduced temperature by 15.23%; increased electrical efficiency by 6.08%
Sonia et al. (2018)	2018	Water circulation	Experimental	Insulated tank with 50 g/s closed-loop flow	Provided sustained cooling and uniform temperature distribution
Basem et al. (2024)	2024	Water-spray cooling	Experimental	Direct water spray on front PV surface	Increased efficiency by 3.9%; stabilized panel temperature at 41 °C
Moraes et al. (2017)	2017	Water flow cooling	Experimental	Front surface water reused with varied flow	Achieved up to 11.99% power gain; cost-effective at lower flow rates
Tashtoush and Al-Oqool (2019)	2019	Water flow cooling	Experimental	Controlled water flow over panel surface	Energy output improved by 15.28–17.75%; optimal operation at 37 °C
Dewi et al. (2024)	2024	Water flow cooling	Experimental	Automated system with microcontroller control	Power output increased by 10.30%; electrical efficiency reached 19.8%
Khan et al. (2023)	2023	Water-spray cooling	Experimental	Single nozzle spray on PV surface	Power increased by 15.3%; temperature dropped from 53 to 23 °C
Mah et al. (2019)	2019	Water-film cooling	Experimental	Water film on rooftop PV; 6 L/min flow rate	Power improved by 15%; energy gain of 0.0178 kWh/h per panel
Murtadha and Hussein (2022)	2022	Nanofluid cooling	Experimental	Water with Al_2O_3 nanofluid at various flow rates	Reduced temperature by 23.14%; power improved by 13% at 3 wt% concentration
Dev et al. (2022)	2022	Nanofluid cooling	Experimental	$Al_2O_3 + TiO_2$ nanofluid at 0.05–0.1 vol. conc	Decreased temperature by 25%; enhanced efficiency by 10.3%

and automated control systems can enhance efficiency with real-time adjustments of water flow. Addressing these challenges will make water-based cooling a more sustainable, efficient, and scalable solution for long-term PV performance. Nanofluids employed in active cooling systems may experience nanoparticle sedimentation, agglomeration, and chemical instability over time. These factors degrade heat transfer performance and may require periodic maintenance or fluid replacement to sustain efficiency. The scarcity of thorough lifecycle assessments underscores the need for future research to assess the sustainability and economic viability of nanofluid-based cooling methods.

While water-based cooling methods significantly enhance photovoltaic (PV) efficiency, their applicability in arid and semi-arid regions raises concerns due to limited water availability. Excessive water use in these environments may compromise the sustainability of such systems, particularly for large-scale deployments. Furthermore, issues such as water evaporation, scarcity, and the need for frequent replenishment increase the operational burden. In response, researchers are exploring alternative water-conserving strategies, such as closed-loop systems that recycle water, fog-based cooling that reduces overall consumption, and hybrid systems combining passive materials like phase change materials (PCMs) or heat sinks to minimize water dependency. Additionally, using reclaimed or non-potable water sources, such as air conditioner condensate, can mitigate water usage without compromising performance. Environmental impact assessments in water-scarce regions must be integrated into the cooling system design to ensure long-term feasibility and sustainability under real-world climate conditions.

2.2.2 Air-based cooling

Air-based cooling is a viable method for increasing the efficiency and lifetime of photovoltaic (PV) panels by exploiting forced air flow for heat removal. Compared to natural convection, this technique provides more thermal control but at the expense of external power supply, which can negate some portion of the efficiency advantage. Forced air cooling systems often include fans, blowers, or heat sinks to improve the heat dissipation, thus resulting in excellent efficiency enhancement.

Recent research has achieved notable efficiency enhancement with air-cooled methods. For instance, utilizing perforated fin heat sinks has been found to improve efficiency by 5.82–5.92%, with non-perforated ones achieving 5.28–5.69% (Hudișteanu et al. 2025). Numerical investigation of air-cooled heat sinks indicated that the optimized fin geometries can achieve a temperature reduction of up to 38.44 °C, leading to improved heat extraction and more stable operation (Al et al. 2025). Experimental arrangements with air ducts incorporating aluminum fins have also suggested higher electrical performance due to elevated heat exchange and PV panel temperature reduction (Kim et al. 2024). Further, an air photovoltaic–thermal collector (PVTC) with longitudinal fins and rectangular turbulators was also found to promote both electrical and thermal performance. The electrical efficiency of the PVTC with fins and turbulators (PVTC-LFRT) was 17.06%, against 16.55% of the smooth channel air PVTC and 16.8% of PVTC with longitudinal fins only. Thermal efficiency increased remarkably to 38.93% against 22.68% and 33.56% of other configurations. Furthermore, the daily total energy output of the PVTC-LFRT was 42.73% higher than that of the smooth air channel PVTC and 11.2% higher than PVTC with only longitudinal fins, confirming its enhanced energy capture and efficiency under real-world conditions (Kim et al. 2024). Figure 8 compares these performance improvements, plotting the outlet air temperature, thermal efficiency and heat gain, power output and electrical efficiency, thermal equivalent efficiency, and daily energy output of the various PVTC configurations. As may be concluded from the results, the PVTC-LFRT system is much superior in both thermal and electrical efficiencies and deserves possible application for enhancing the energy conversion efficiency of the PV panels.

For comparative studies, researchers have demonstrated that forced air cooling can drop the temperature of PV panels by 9 °C, enhancing electrical efficiency by 7–12.6% (Patil et al. 2023). In another investigation of Jet-in-Crossflow (JICF) cooling, it was found that impingement jets at the back of PV panels decreased temperature by 8–21 °C and efficiency by 4–11% (Ali et al. 2025a). Likewise, air-to-air heat exchangers with blowers reduced module temperatures from 40 to 34 °C and improved efficiency by 2% and power output by 12.8% (Abushgair 2021). The results show that the insertion of fins and turbulators in air cooling systems not only allows for achieving high thermal efficiency but also increased electrical output, thereby rendering the method an option for the optimization of PV performance in hot climate conditions.

Though forced air cooling does need an external power source, its capacity to keep the panel temperature lower and enhance energy production is higher than the power it uses in most situations. Besides, the application of passive air-cooling methods involving air, like improved fin designs and augmented flow channels, is an environmentally friendly thermal management method without the additional energy cost. Further work must be directed towards optimization of these methods, investigating hybrid air and phase change material (PCM) cooling systems, and determining the long-term economic feasibility of large-scale use in various climatic conditions. As airflow dynamics improve and there is optimization of the cooling process, air-based cooling has the potential to play a large role in the overall efficiency and sustainability of photovoltaic energy systems.

2.2.3 Jet impingement cooling

Jet impingement cooling is an effective thermal management method that employs high-velocity air or fluid jets to cool the PV panels. The technique improves the heat transfer rates, which leads to excellent temperature decrease and electrical efficiency (Ali et al. 2025b). Jet impingement is especially advantageous over passive and traditional cooling techniques in ensuring uniform temperature distribution on the PV surface, thus reducing thermal stresses and hotspots that tend to lower the performance of panels over time (Javidan and Moghadam 2021; Suja et al. 2023). Despite its high efficiency, jet impingement cooling is comparatively complicated and expensive, requiring intricate optimization of nozzle geometries, flow rates, and coolants to achieve optimum efficiency. Figure 9a and b present experimental results where jet impingement significantly reduced PV temperatures from 69.7 to 36.6 °C in summer and 47.6 to 31.1 °C in winter, with an efficiency improvement of 51.6%. This demonstrates the effectiveness of the technique in managing temperature and enhancing PV panel performance (Bahaidarah 2016a). Another study using nanofluid jets achieved cooling down to 36.4 °C, showcasing that next-generation coolants offer superior cooling performance, as shown in Fig. 9c (Mohammadpour et al. 2022). Furthermore, as demonstrated in Fig. 9d, the combination of jet impingement cooling with compound parabolic concentrators (CPC) improved electrical efficiency by 7% and power output by 36% (Masood et al. 2022).

In comparison with other cooling techniques, jet impingement possesses certain exclusive benefits. Whereas front

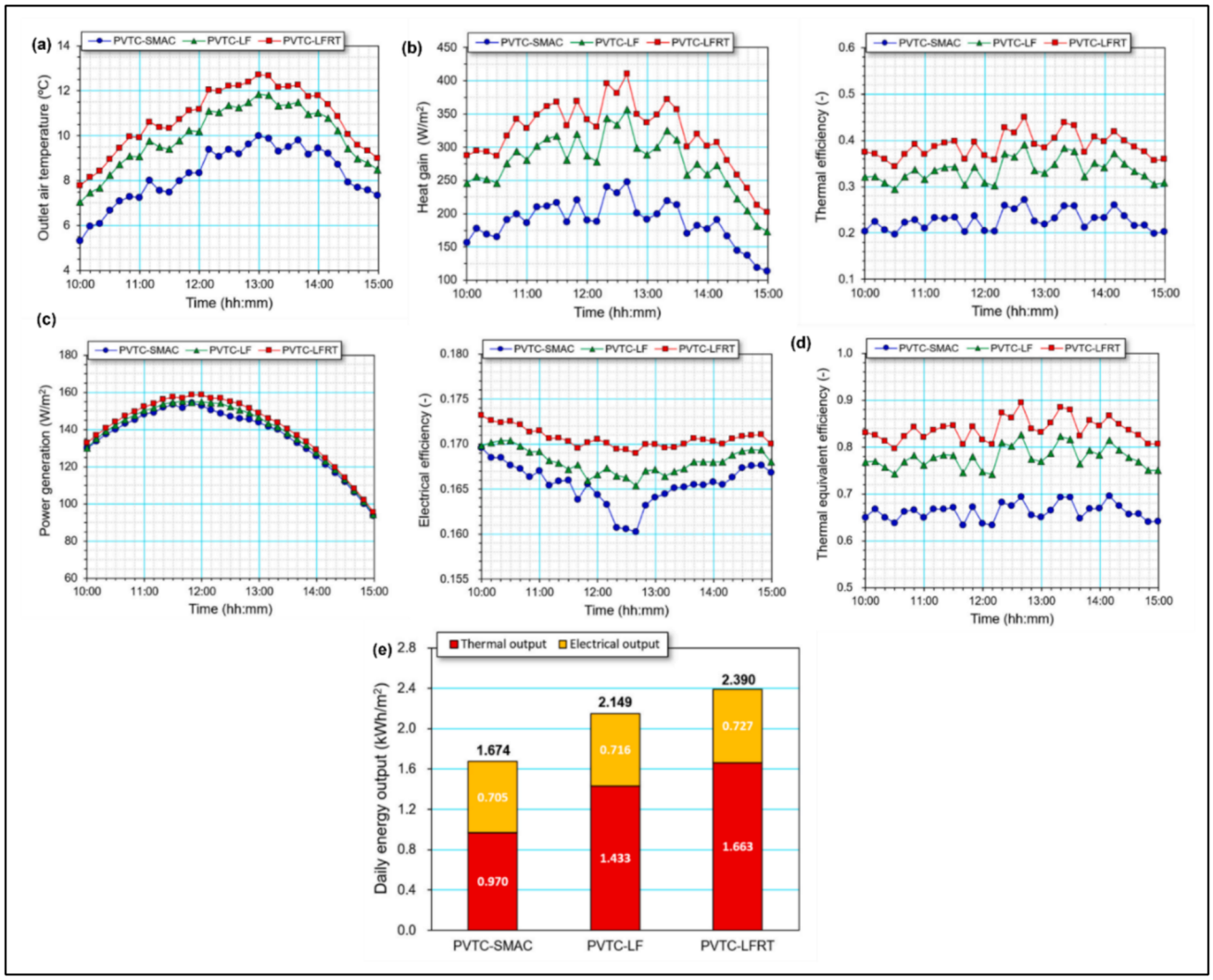


Fig. 8 **a** Outlet air temperature of different PVTCs, **b** Heat gain and thermal efficiency of different PVTCs, **c** Power generation and electrical efficiency of different PVTCs, **d** Thermal equivalent efficiency of different PVTCs, **e** Daily energy output (thermal and electrical) of different PVTCs (Kim et al. 2024)

water cooling enhances efficiency by 9%, jet cooling attains as much as 51.6% (Bahaidarah 2016b; Bhat et al. 2022). Hybrid designs integrating jet impingement with microchannels have additionally improved cooling, reducing the temperature of PV cells to 55 °C and increasing efficiency to 39.7% (Abo-Zahhad et al. 2019). The efficiency of jet cooling relies on parameters like the location of nozzles, type of coolant, and flow rate. Nozzle configurations that are optimized provide uniform cooling (Plant et al. 2023), and nanofluids provide higher thermal conductivity compared to air or water (Suja et al. 2023). However, increased flow rates improve cooling at the expense of increased energy consumption (Gupta et al. 2023).

Despite significant advantages in temperature reduction and efficiency improvement, active cooling methods—including jet impingement cooling—face challenges that limit their broader adoption. Jet impingement cooling, in particular, involves high installation costs, energy consumption, and system complexity due to the need for precise nozzle design, controlled flow rates, and advanced coolants. More generally, many active cooling studies are conducted under controlled laboratory conditions with limited sample sizes and short durations, which may not fully represent real-world variability or long-term stability. Differences in testing setups, coolant types, and configurations also complicate result comparisons. Furthermore, the energy consumption of pumps, fans, and other components must be carefully evaluated to ensure net efficiency gains, a factor that is not

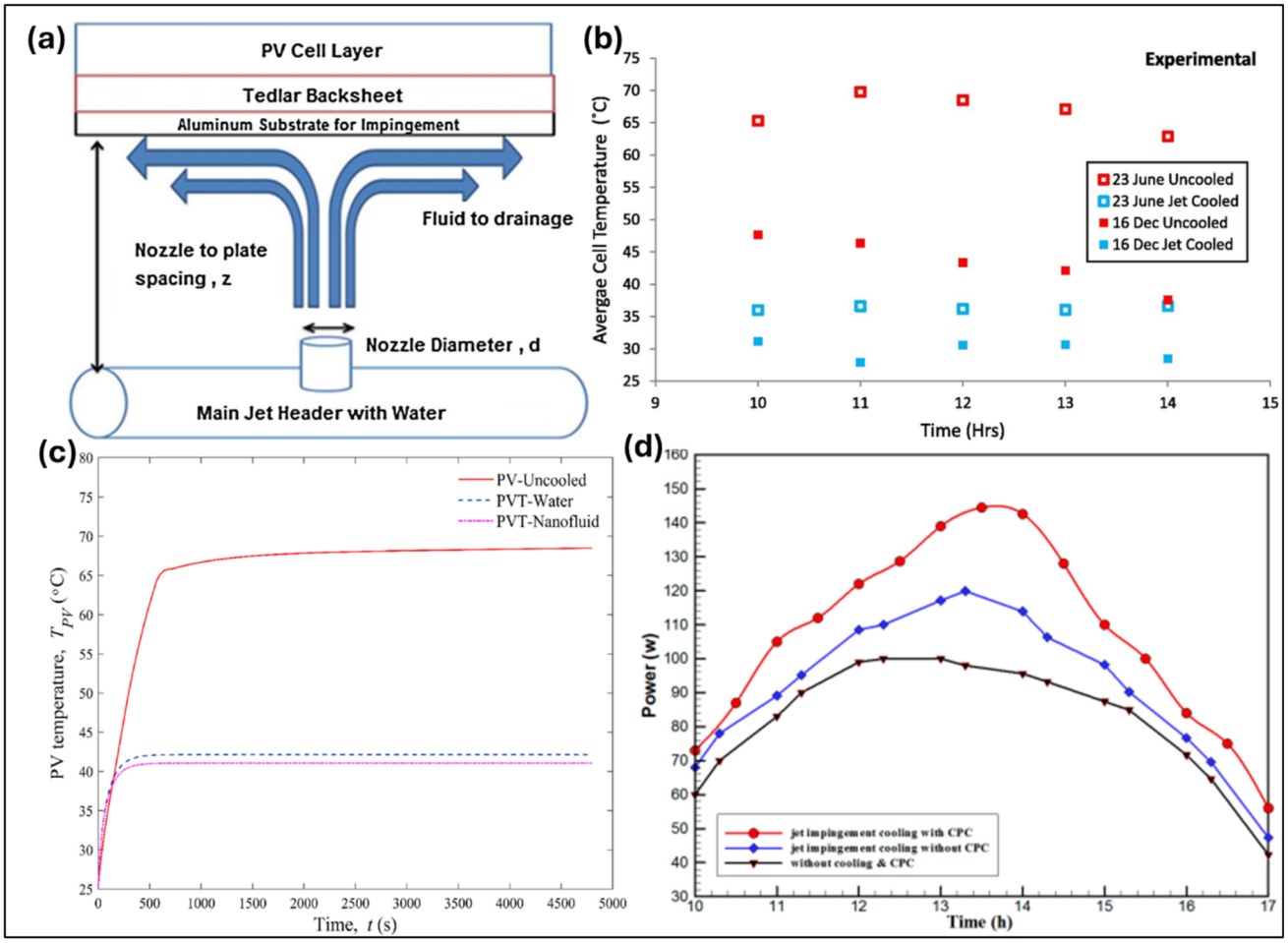


Fig. 9 Experimental setup and performance results of jet impingement cooling for PV panels. **a** Schematic layout of the jet impingement cooling system, showing the nozzle arrangement, cooling fluid direction, and drainage setup. This illustrates the mechanism through which fluid jets target the rear surface of the PV panel to dissipate heat. **b** Experimental comparison of PV surface temperatures with and without jet cooling during summer and winter. Results show a reduction from 69.7 °C to 36.6 °C in summer and from 47.6 °C to 31.1 °C in winter, confirming the high efficiency of jet cooling under varying climatic conditions. Adapted from (Bahaidarah 2016a). **c** Time-series temperature profiles for different cooling methods, including nanofluids. The graph demonstrates that jet impingement with nanofluids provides the lowest steady-state temperature (36.4 °C), outperforming conventional techniques. Adapted from (Mohammadpour et al. 2022). **d** Electrical power output comparison for standard PV and PV with compound parabolic concentrator (CPC) and jet cooling. Results show a 36% increase in power output, validating the combined effect of optical concentration and thermal regulation. Adapted from Hameed Jaaz et al. (2017)

always thoroughly reported. Integration of active cooling into commercial photovoltaic systems is challenged by system complexity, maintenance demands, and upfront costs. To address these limitations, advancements in hybrid cooling designs, AI-driven optimization, and energy-efficient components show promise. However, comprehensive field-scale trials, lifecycle assessments, and standardized testing protocols remain essential to robustly validate the performance, economic viability, and sustainability of active cooling technologies prior to widespread commercial deployment.

To facilitate the comparison of various cooling methods, Table 5 provides a consolidated summary outlining their temperature reduction capacity, electrical efficiency improvement, water requirements, cost implications, AI integration potential, and suitability across different climatic and operational contexts. This comparative overview enables informed decision-making regarding the deployment of cooling technologies in photovoltaic systems under real-world conditions.

This comparison illustrates that while jet impingement and water-based cooling achieve the highest temperature reduction and efficiency gains, their practical adoption is constrained by cost and water availability, respectively. In contrast, radiative and PCM cooling offer promising low-maintenance options for dry or off-grid regions. The

Table 5 Summary of PV cooling techniques across key criteria

Cooling method	Temp. reduction (°C)	Efficiency gain (%)	Water demand	Cost	AI integration potential	Best suited conditions
Radiative cooling	3–10.6	1.35–16.8	None	Low	High	Arid, clear-sky, low-dust areas
PCM cooling	3–37	1.3–30	None	Moderate	Medium	High irradiance, stable diurnal range
Natural convection	2.7–12.7	1.2–5.92	None	Low	Low	Moderate climate, rooftop setups
Water-based cooling	10–30 +	3.9–22.72	High	Moderate-high	Medium	High solar flux, water-available zones
Air-based cooling	9–38.4	5.28–12.6	None	Moderate	Medium	Industrial areas, low humidity zones
Jet impingement cooling	8–33.1	7–51.6	Variable	High	High	Research sites, hybrid system trials

integration of AI-driven control systems may further enhance the viability of these methods in diverse operating conditions.

2.3 Hybrid cooling strategies

Hybrid cooling strategies combine passive (such as phase change materials [PCMs], heat pipes) and active (like water/air circulation) methods to enhance the efficiency and lifespan of PV panels by managing their operating temperature. Systems such as PV/solar air heaters (PV/SAH) and PV/thermal-radiative cooling (PV/T-RC) integrate these methods to address performance challenges caused by heat buildup (Farzan et al. 2024; Dixit et al. 2023). Emerging hybrid photovoltaic systems, such as thermophotovoltaic (TPV) and photovoltaic-thermal (PVT) hybrids, represent advanced approaches to thermal management and energy harvesting. TPV systems convert thermal radiation into electricity and require effective heat dissipation to maximize efficiency and power output. PVT hybrids integrate electrical and thermal energy production, employing cooling techniques like liquid cooling, phase change materials, and nanofluids to reduce PV temperatures and enhance overall performance (Hachicha et al. 2022; Togun et al. 2025; Roux et al. 2024; Kumar and Dhiman 2023). These technologies broaden the scope of hybrid cooling strategies by combining thermal and electrical optimization, achieving significant temperature reductions and efficiency gains. These strategies reduce PV temperatures and improve energy efficiency, with temperature reductions up to 14.7 °C and exergy efficiency improvements of up to 8.71% (Kumar et al. 2024a). Additionally, integrating fluid filters with active cooling mechanisms can increase system efficiency by over 38% (Otanicar et al. 2011), thus optimizing energy output and extending the operational lifespan of PV panels (Ali et al. 2024a). Azmi et al. (2023) demonstrated the effectiveness of a hybrid air–water cooling system, which improved PV efficiency by approximately 4.5% compared to water-only cooling.

Singh et al. (2022) explored the performance of PV modules under hybrid cooling methods combining thermoelectric coolers (TECs) and PCMs, achieving a 9.7% efficiency increase with the combination of Vaseline as PCM and silver as the back casing. Their hybrid system, incorporating TECs, PCM, and heat sinks, resulted in a 19.4% increase in electrical efficiency and 19.32% in panel efficiency. These results are visually represented in Fig. 10a through e, which illustrate the hybrid system's performance improvements, including the configuration, temperature reduction over time, and efficiency increments. The figures clearly demonstrate the enhanced cooling effectiveness and efficiency gains achieved by combining TECs and PCMs.

Recent advancements in artificial intelligence (AI) have also emerged as powerful tools in optimizing hybrid PV cooling systems. AI-driven models, particularly machine learning algorithms, can predict environmental and operating conditions, enabling adaptive control of hybrid components such as thermoelectric coolers, nanofluid flows, and PCMs. For instance, predictive thermal modeling has demonstrated that AI-based controllers can reduce PV surface temperature by up to 6.7% and increase electrical output by approximately 10%, depending on the dynamic optimization of fluid flow and cooling mode transitions. Moreover, real-time AI-assisted systems can analyze solar irradiance,

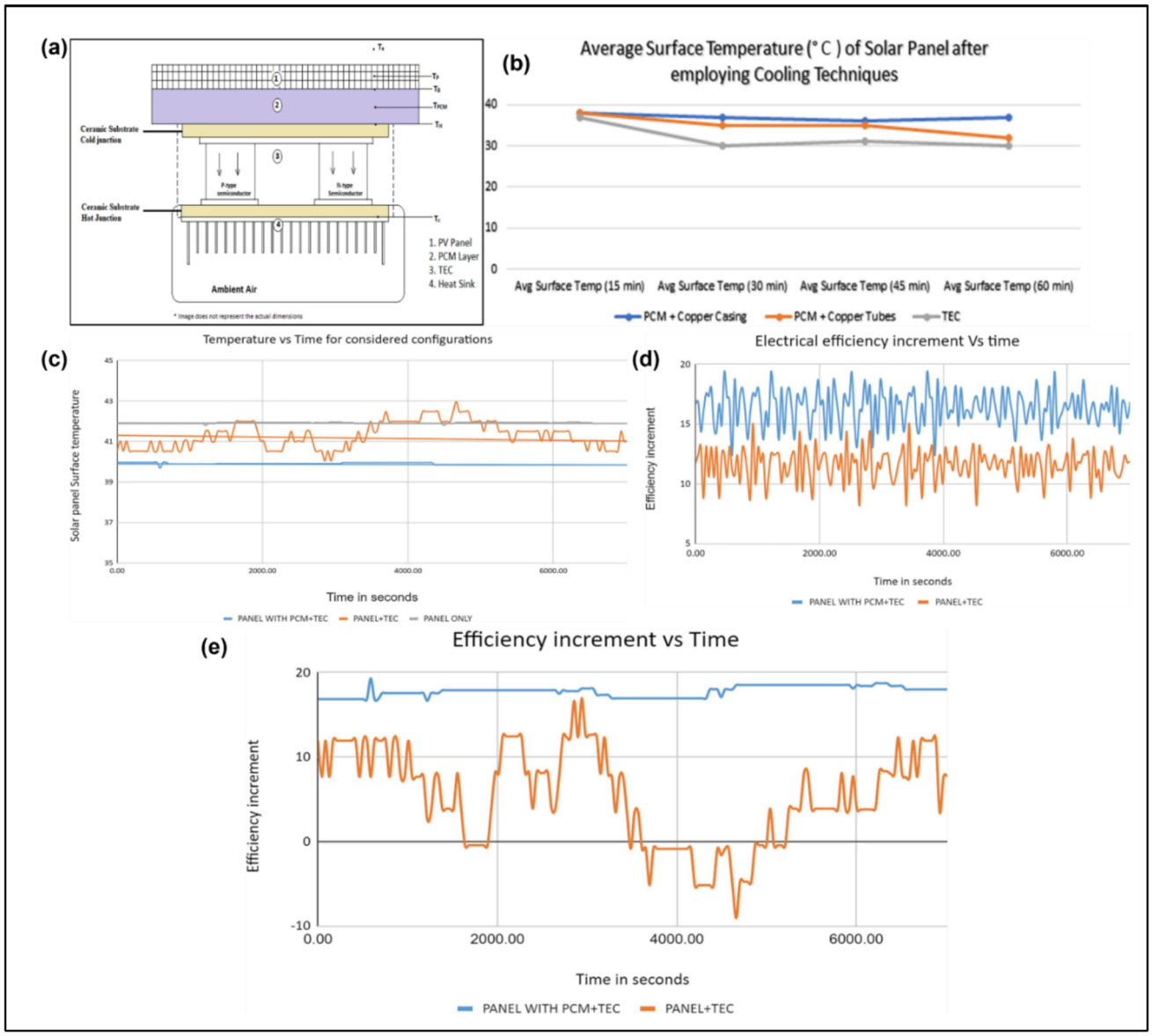


Fig. 10 Hybrid cooling system for PV panels with thermoelectric coolers (TECs), phase change materials (PCMs), and heat sinks. **a** Diagram of the hybrid model using TECs, PCMs, and heat sinks for cooling. **b** Average surface temperature of the PV panel at intervals of 15 min for different cooling techniques. **c** Temperature vs. time for the PV panel under various cooling configurations. **d** Increment in electrical efficiency of the PV panel over time for the hybrid cooling model. **e** Efficiency increment vs. time for the PV panel using the hybrid cooling model (Reproduced from Singh et al. 2022)

ambient temperature, and load demands to autonomously switch between passive and active cooling modes, ensuring efficient thermal regulation with minimal energy input. Integrating AI with hybrid cooling systems thus strengthens the synergy between mechanical configuration and intelligent control, paving the way for more responsive, self-regulating, and energy-efficient photovoltaic installations. While many studies focus on first-law performance indicators—such as temperature drop and electrical efficiency—there is growing recognition of the importance of second-law thermodynamic metrics in evaluating hybrid PV systems. Exergy analysis measures the maximum useful work a system can deliver, identifying inefficiencies due to irreversibility. Entropy generation, in turn, quantifies these irreversibilities and reflects the degree of energy degradation. Including these metrics offers deeper insight into the thermodynamic quality of energy transformations, especially in complex systems where multiple cooling strategies interact.

Several experimental studies have demonstrated the effectiveness of hybrid PV cooling systems from a second-law perspective. Alktranee et al. (2024) showed that Fe_2O_3 nanofluids improved exergy efficiency by 13% and reduced

entropy generation by 68%. TiO_2–CuO hybrid nanofluids yielded even more impressive results, with a 14.97% boost in exergy efficiency and nearly 70% reduction in entropy generation (Alktranee et al. 2023a). Similarly, ZrO_2 nanofluids improved exergy efficiency by 66.8% and reduced entropy generation by 26% (Alktranee et al. 2023b). System enhancements such as twisted tape inserts have also shown to improve exergy efficiency by up to 14.86% in indoor experimental setups (Kalateh et al. 2022), while porous metal foams and hybrid cooling geometries further help reduce entropy production and enhance thermodynamic performance (Nedjem et al. 2024; Alsharifi et al. 2024).

However, a major limitation in applying second-law analysis is the inconsistent reporting of critical thermophysical parameters across the literature. Many experimental studies fail to document essential data such as mass flow rates, inlet and outlet temperatures, ambient reference conditions, and specific heat capacities. This lack of standardized reporting makes it difficult to reliably calculate and compare exergy efficiency and entropy generation across different studies. Consequently, cross-comparative assessments may be inaccurate or incomplete, limiting the ability to draw meaningful conclusions about the true thermodynamic performance of various cooling strategies. This challenge has also been highlighted by researchers such as (Arshi Banu et al. 2020), who emphasized the critical need for standardized reporting of thermophysical data to support more reliable and comprehensive second-law analyses across studies.

Therefore, while this review primarily emphasizes first-law metrics due to data availability, second-law considerations are indispensable for understanding the true thermodynamic efficiency of PV cooling systems. Future studies should prioritize the inclusion of second-law parameters in their performance reports to enable meaningful comparative analysis and support the development of more energy-efficient and sustainable hybrid cooling technologies.

2.4 Normalized benchmarking framework for PV cooling strategies

To develop a normalized benchmarking framework for evaluating photovoltaic (PV) cooling strategies, it is essential to consider key variables such as irradiance, module type, and ambient temperature, which directly affect the efficiency of cooling methods. Cooling strategies can significantly improve PV performance by lowering temperatures and enhancing electrical efficiency, but their effectiveness can vary based on these variables. For instance, immersion cooling, which utilizes liquids like dimethyl silicone oil, has been shown to minimize power loss and maintain high transmittance (Han et al. 2012, 2018a, b). In contrast, paraffin wax, when used in aluminum cans, has demonstrated a notable improvement in electrical efficiency, increasing from 10.69 to 12.60%, while boosting power output efficiency from 61.72 to 71.56% (Tuncer et al. 2022). Similarly, latent heat storage systems (LHSS), which use metallic mesh layers and nanoparticles, have proven effective in reducing surface temperature and improving module efficiency (Gürbüz et al. 2024). Furthermore, nanofluid cooling, such as Ag/water nanofluids, enhances both energy and exergy efficiencies, leading to a 35% increase in power output (Aberoumand et al. 2018). Other cooling techniques like water flow on PV modules have been observed to reduce temperature and increase energy output by about 30%, while hybrid cooling, combining water spray and air cooling, has shown a reduction in panel temperature by up to 40%, with a 13% increase in output power (Al-Masalha et al. 2024; Ramgolam et al. 2021).

In terms of irradiance, studies often employ both controlled and natural conditions to assess the cooling efficacy of various methods. Controlled irradiance, typically provided by solar simulators, allows for a consistent evaluation of cooling methods, whereas natural irradiance, especially in tropical climates, offers real-world conditions that can highlight the challenges and benefits of different strategies (Racmanto et al. 2024). Module types also play a critical role, with triple-junction solar cells benefiting from immersion cooling due to their high concentration PV systems (Han et al. 2018a, b). On the other hand, conventional PV panels are more commonly studied using methods like paraffin wax and nanofluid cooling (Tuncer et al. 2022; Aberoumand et al. 2018). Ambient temperature is another influential factor, particularly in hot climates where cooling methods are most needed to prevent overheating. For example, hybrid cooling systems can reduce panel temperatures by as much as 40%, which translates to increased performance in high-temperature environments (Al-Masalha et al. 2024).

The effectiveness of photovoltaic cooling strategies varies significantly across different climatic zones, highlighting the necessity to consider local environmental conditions in benchmarking frameworks. In arid climates, water cooling, air cooling, and hybrid air–water systems are highly effective, achieving temperature reductions up to 24 °C and improving output power by 13% (Al-Masalha et al. 2024; Akther et al. 2024). Nighttime radiative cooling also enhances efficiency by lowering PV cell temperatures by 2–7 °C (Zaite et al. 2022). However, high temperatures above 40 °C present performance challenges that cooling must mitigate (Tursunov et al. 2025). In humid regions, evaporative cooling methods such as absorbent pads and synthetic clays work well but consume considerable water and face diminished effectiveness due to high relative humidity (Zaite et al. 2022; Cengiz et al. 2024). Tropical climates benefit from phase change materials (PCMs) combined with thermal insulation and photovoltaic-thermal (PV/T) systems, which improve electrical efficiencies significantly—from 12.53 to

29.06%—while reducing cooling energy demands by around 12% (Nematchoua et al. 2020; do Nascimento et al. 2022). These climatic differences must be integrated into any normalized benchmarking framework to select cooling methods optimized for specific environmental conditions, thus ensuring realistic efficiency improvements and system reliability.

To enable direct comparisons across various cooling techniques, the normalized efficiency of each method must be considered. For example, paraffin wax cooling has shown an increase in efficiency from 61.72 to 71.56% (Tuncer et al. 2022), while nanofluid cooling has resulted in a 35% increase in power output (Aberoumand et al. 2018), and hybrid cooling has achieved up to a 13% improvement in power (Al-Masalha et al. 2024). By considering these factors in a normalized framework, a clearer picture emerges of which cooling methods are most effective under varying conditions, allowing researchers and engineers to select the optimal strategies for improving PV efficiency.

3 Performance comparison of cooling strategies

Cooling techniques are compared to each other based on the degree of photovoltaic (PV) performance enhancement, temperature reduction, and efficiency improvement. Relative evaluation of various cooling techniques, including temperature reduction, efficiency improvement, economic feasibility, environmental impact, and suitability in various climates, is presented in this section.

3.1 Effectiveness in temperature reduction

PV panel performance is highly sensitive to temperature, with efficiency decreasing by 0.4–0.5% per 1 °C rise in temperature (Koohestani et al. 2023; Aljuwaysir Salah Fahad et al. 2025). Extended heat exposure increases the thermal degradation rate, decreasing lifespan and reliability (Sultan et al. 2021; Arcuri et al. 2014). Ambient temperature, solar irradiance, and wind speed are climatic factors also influencing performance, and cooling strategies are necessary for long-term energy outputs (Homa et al. 2024). Cooling methods are referred to as passive, active, and hybrid with varying efficiency depending on ambient conditions and system complexity. Passive cooling such as radiative cooling, phase change material (PCM), and natural convection lowers the temperature by 6–20 °C, enhancing efficiency up to 15.5% (Sargunanathan et al. 2016; Sharma et al. 2018). Radiative cooling using PCM is highly efficient in dry hot climates but restricted during the daytime because of lower thermal radiation (Nižetić et al. 2018). Heat sinks, aluminum fins, and heat pipes also facilitate cooling but require airflow and installation design.

Active cooling techniques, including water flow, forced air, and nanofluid immersion, provide greater temperature drops (30–55 °C) and efficiency improvement up to 22.2% (Libra et al. 2022; Ye et al. 2009). Water cooling, in particular evaporative spray systems, is efficient but not viable for the desert given its aqueous nature (Ibrahim et al. 2024). Increasingly effective concentrated PV cooling is more expensive. Hybrid cooling merges passive with active systems for the best performance, reducing temperature by as much as 40 °C and boosting efficiency by 30% (Yazdanifard et al. 2021; Vaillon et al. 2018). PCM-nanofluid incorporation provides assurance of thermal reliability and hence hybrids are the best for extreme climates. AI-based optimization also boosts efficiency by modifying cooling methods in real-time as the situation evolves. Cooling techniques, performance, and constraints are highlighted in Table 6.

The best PV cooling method is chosen based on climatic conditions, availability of water resources, and complexity of the system. Passive cooling systems represent low-maintenance and low-cost solutions with diminishing efficiency in extremely hot climates. Active cooling systems, including water-based cooling and forced air systems, are capable of significant temperature reduction at the expense of increased resource consumption. Meanwhile, hybrid cooling techniques, which integrate passive and active mechanisms, exhibit the highest efficiency gains and are optimally applicable to high solar irradiance locations.

3.2 Impact on PV efficiency improvement

PV efficiency drops by 0.4–0.5% with every 1 °C increase due to decreased open-circuit voltage and higher recombination losses (Rahman et al. 2017; Farahat 2004). Cooling reduces these losses, resulting in improved performance and longer PV life (Basem et al. 2024; Awad et al. 2022a). Cooling techniques differ with improvements in efficiency, as shown in Fig. 11. Water cooling produces 1.6–10.3% efficiency enhancement (Amr et al. 2024; Basem et al. 2024; Dewi et al. 2024), whereas nanofluid cooling results in larger percentages, e.g., Al_2O_3-water 13% and MWCNT/ZnO 16.8% (Basem et al. 2024; Dev et al. 2022; Sharaby et al. 2024). Air cooling provides smaller but decent increments (1.1–4%) (Basem et al. 2024; Almuwailhi and Zeitoun 2023). PCM cooling maintains PV temperature stable and its effect on the impact is a function of material properties (Siah Chehreh Ghadikolaei 2021). Hybrid and advanced techniques yield the highest improvements, with PV/T-nanofluid systems increasing efficiency by 10.3%, radiative cooling with heat sinks by 11.11%, and active PCM cooling by 19.32% (Kumar et al. 2023a; Dev et al. 2022; Wang et al. 2019). Selecting the best cooling method requires balancing efficiency, feasibility, and environmental constraints. Water cooling, although efficient, is impractical for dry climates owing to water demand

Table 6 Temperature reduction performance of different PV cooling methods in various configurations

Refs.	Cooling method	Cooling mechanism	Temperature reduction (°C)	Limitations
Lamba et al. (2024)	Heat sink + radiative cooling	Passive	22 °C	Complex integration, location-specific PCM selection
Amr et al. (2024)	Evaporative cooling (nozzles)	Active	44.8% reduction	High water usage, maintenance
Safar et al. (2020)	Active cooling	Active	33 °C	Energy consumption for active cooling
Jaleel Mahdi et al. (2024)	Ground source cooling	Passive	28% reduction	Installation complexity, location-specific
Chavan and Devaprakasam (2019)	Water-based PVT-PCM	Hybrid	8.10% reduction	Requires water supply, PCM management
Kumar et al. (2024a)	Heat pipes	Passive	11 °C	Installation complexity
Popovici et al. (2016)	Air cooled heat sinks	Passive	Not specified	Efficiency dependent on air flow configuration
Schiro et al. (2016)	Water spray cooling	Active	Not specified	Water usage, potential for water wastage
Kaneesamkandi and Rehman (2023)	Aluminum fin cooling	Passive	Not specified	Installation complexity
Yousef et al. (2016)	Concentrated PV cooling	Active	25–30% reduction	High initial cost, maintenance
Baz et al. (2024)	Water distribution cooling	Active	30–35% reduction	Water usage, pressure drop issues
Nabil and Mansour (2022)	Front surface water cooling	Active	29.37% reduction	Water usage, maintenance
Kumar et al. (2023b)	Heat pipe + radiative cooling	Passive	13 °C	Complex integration, location-specific
Kumar et al. (2023a)	Heat sink + radiative cooling	Passive	6.63% reduction	Complex integration

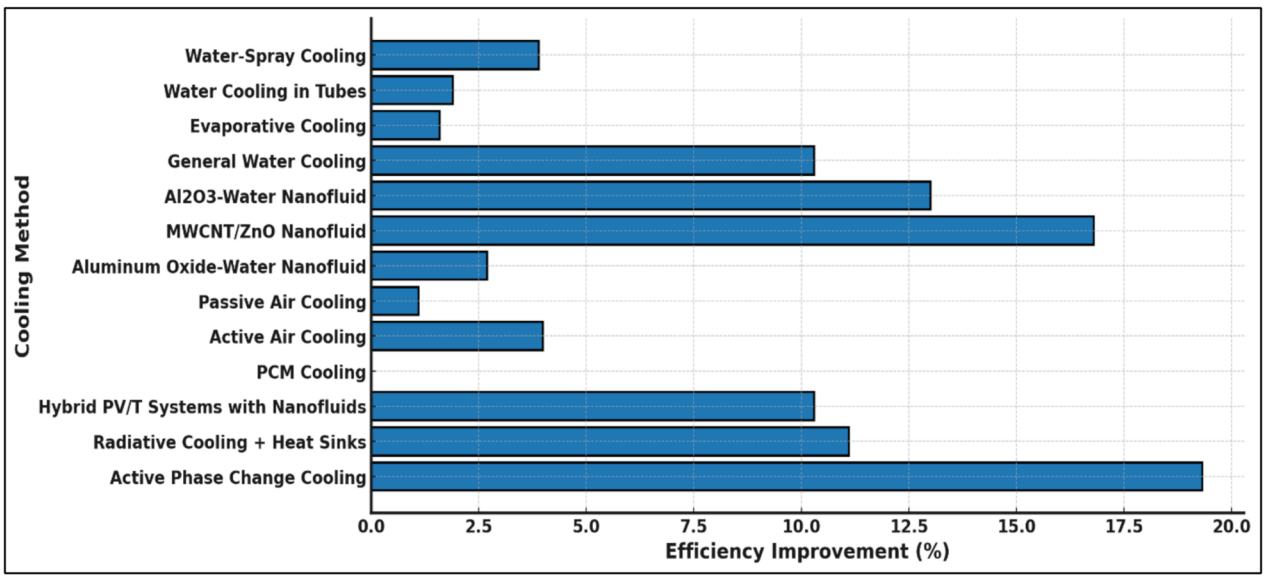


Fig. 11 Comparison of cooling methods for PV efficiency improvement

(Basem et al. 2024). PCM cooling enhances long-term performance but deteriorates with time (Sen and Celiktas 2024; Hasan et al. 2017). AI-based cooling enhances performance dynamically at the expense of system complexity and cost (Koohestani et al. 2023; Aljuwaysir Salah Fahad et al. 2025). These trade-offs highlight the need for tailored cooling solutions based on climate, cost, and performance goals.

3.3 Economic and environmental feasibility

3.3.1 Economic feasibility

The economic feasibility of PV panel cooling technology is based on initial expenditure, future savings, and recurring expenditure, with Levelized Cost of Energy (LCOE) and Net Present Value (NPV) being the indicators of feasibility, which guide decision-making. AI-based cooling technology has high initial expenses in the form of infrastructure and sensor integration but comes with high future savings in terms of efficient usage of energy and minimizing thermal losses (Akrouch et al. 2024; Nižetić et al. 2018). Phase change materials (PCM), on the other hand, are cheap initially but degrade over time and must be replaced from time to time to keep the efficiency up (Čorić et al. 2024). Water cooling is medium to high initial cost, enhancing PV efficiency but needing system optimization for it to be cost-effective (Castanheira et al. 2018). Hybrid systems with passive and active cooling have better performance but need careful cost–benefit analysis to compensate for the increased initial costs (Elsabahy et al. 2022).

Maintenance cost significantly affects the feasibility of such cooling strategies, especially for large-scale PV systems. Systems based on water are susceptible to frequent cleaning and refilling, thus increasing operating costs and restricting feasibility in desert areas (Nižetić et al. 2017). Passive cooling methods, such as heat sinks and natural convection, are more cost-effective due to minimal maintenance requirements and long-term reliability (Chandel and Agarwal 2017). However, hybrid systems, despite their higher initial and maintenance costs, offer the best return on investment by combining the strengths of both passive and active cooling (Ali 2020). Given the trade-off between cost and efficiency, AI-driven cooling provides the highest long-term economic benefits, while hybrid methods balance cost-effectiveness and energy savings, making them viable for large-scale PV deployment (Tina et al. 2023).

3.3.2 Environmental impact

The feasibility of PV cooling depends on resource availability, environmental impact, and material degradation in extensive applications. Water cooling, as efficient as it may be, is not sustainable in arid regions due to excessive consumption and evaporation losses (Abdul Rahim et al. 2024). Air cooling is cost-effective but underperforms in hot climates where passive cooling is ineffective (Aljuwaysir Salah Fahad et al. 2025). While PCMs regulate temperature effectively, they are plagued by degradation, thus worsening waste management problems in extensive farms (Chandel and Agarwal 2017). Nanofluids improve heat transfer but are environmentally hazardous as nanoparticles escape into soil and aquatic food chains is possible (Mishra et al. 2023).

By comparing environmental issues, PCMs provide an environmentally benign but temporary solution by means of degradation, whereas nanofluids enhance cooling potential but at the expense of rigorous containment lest they pollute (Abdelrazik et al. 2018). Water cooling is efficient but conflicts with vital water needs in arid areas (Castanheira et al. 2018). Air cooling, of minimal environmental effect, is best adapted to temperate climates (Koohestani et al. 2023). Based on these trade-offs, it is suggested that future studies explore hybrid cooling systems combining passive and active approaches with the help of AI-powered optimization for improved efficiency and sustainability (Tina et al. 2023).

3.4 Trade-offs and suitability in different environments

PV cooling techniques possess climate-dependent efficiency. Evaporative cooling techniques such as synthetic clay and absorbent pads improve heat sinking in dry, hot climates, boosting the efficiency of PV by as much as 20% (Cengiz et al. 2024). Thermoelectric cooling, powered by solar energy, also provides efficient thermal management in these areas (Irshad et al. 2024). Night radiative cooling does not work in humid climates because of the high humidity, but PCMs stabilize PV temperatures and are therefore applicable in tropical climates (Metwally et al. 2021).

Scalability determines cooling selection for large PV farms and small rooftop systems. Artificial intelligence-based cooling, via machine learning capable of forecasting thermal changes and reacting thereto, is applicable for large plants, optimally balancing heat loads (Kaneesamkandi and Rehman 2023). Thermosiphon cooling is an inexpensive, scalable solution for large PV farms (Dirawi et al. 2025), while passive PCM-based cooling is most suitable for small rooftop systems owing to its energy-free nature and ease of implementation (Sheikh et al. 2024b). Big farms enjoy economies of scale though are constrained by land use, energy storage, and maintenance requirements, while smaller systems provide greater flexibility for decentralized cooling (Sheikh et al. 2024b). Table 7 provides an overview of the suitability of the different cooling techniques to varying environmental conditions.

Table 7 Comparison of photovoltaic (PV) cooling methods based on environmental suitability

Refs.	Cooling method	Best for arid regions?	Best for humid climates?	Best for high-sun exposure?	Effective at night?
Cengiz et al. (2024) and Lamba et al. (2024)	Phase change materials (PCM)	✓	X	✓	X
Sun et al. (2024)	Radiative cooling (RC)	✓	✓	✓	✓
Lamba et al. (2024)	Heat sink (HS)	✓	X	✓	X
Amr et al. (2024)	Evaporative cooling	✓	X	✓	X
Nabil and Mansour (2022) and Patel et al. (2023)	Water cooling	✓	✓	✓	X
Amr et al. (2024) and Nabil and Mansour (2022)	Air cooling	✓	✓	✓	X
Kaneesamkandi and Rehman (2023)	Thermosiphon cooling	✓	✓	✓	X
Aljuwaysir Salah Fahad et al. (2025)	Forced convection cooling	✓	✓	✓	X
Geng et al. (2012)	Heat pipe	✓	✓	✓	X

The applicability and compromises of these cooling techniques are a function of environment, cost, and system size. As indicated in Table 5, while radiative cooling is most versatile under any environment, evaporative cooling and PCM are most appropriate for dry environments with plenty of sunshine but not so suitable in humid environments. Water- and air-cooling systems are both most suitable for dry and humid weather but are inefficient at night. AI-cooling and thermosiphon cooling provide scalable systems and thus are best suited for large solar plants. Such comparisons ensure optimum PV system performance according to location-specific needs.

4 AI and smart control in photovoltaic (PV) cooling

Artificial intelligence and smart control systems maximize PV cooling through the prediction of temperature changes, automatization of the response, and efficiency enhancement. This subsection discusses how AI-based cooling techniques enhance PV performance through machine learning, hybrid cooling optimization, and adaptive control in real time.

4.1 Role of AI in photovoltaic cooling

AI-driven cooling systems maximize PV thermal management compared to conventional methods through real-time monitoring, predictive analytics, and adaptive control. AI adjusts cooling parameters in real time according to ambient conditions rather than passive cooling through natural convection or time-based active cooling. For instance, reinforcement learning (RL) has been utilized to autonomously optimize cooling strategies, where the model continuously adapts to changing environmental conditions to maximize efficiency. Similarly, Long Short-Term Memory (LSTM) networks have been applied to predict temperature fluctuations and solar irradiance patterns, enabling more accurate cooling system adjustments.

It is important to recognize that transient thermal behaviors, such as dawn and dusk thermal lag, play a significant role in PV panel temperature dynamics and cooling performance. These transient effects arise from the gradual changes in solar irradiance and ambient conditions during sunrise and sunset, causing a delay between environmental temperature changes and panel temperature response. Most existing cooling strategies and AI models tend to assume steady-state conditions, which may overlook these diurnal variations and lead to suboptimal cooling performance under real-world daily cycles. Incorporating transient thermal dynamics into cooling models and AI algorithms is therefore essential to ensure accurate predictions and effective temperature management throughout the entire day, ultimately enhancing PV system efficiency and longevity (Chan-Dzib et al. 2025; Heusinger et al. 2020; Darian and Ghorreshi 2021).

One example of AI optimization in a PV/T system with Optimized Deep Neural Network (ODNN) and Sand Cat Swarm Optimization (SCSO) achieved high prediction accuracy ($R^2 = 0.9972$, MSE $= 0.00251$), improving energy efficiency and cost savings (Shaban et al. 2024). These models, trained on large datasets of real-time temperature, irradiance, and weather data from PV systems, achieve robust

prediction performance, avoiding overfitting by incorporating techniques like cross-validation and regularization. Similarly, the Infer-Cool system reduced GPU temperature by 5 °C and achieved 20% cool energy savings, demonstrating AI's ability to minimize energy waste and enhance performance (Pei et al. 2024). Although active cooling is generally 22.2% more effective than passive cooling, AI-based methods attain comparable or better performance through the utilization of weather prediction and dynamic control (Safar et al. 2020).

Beyond real-time adjustments, AI models predict PV temperature trends and proactively optimize cooling strategies, reducing thermal stress and energy consumption. Proactive cooling is achieved through more sophisticated systems such as Gradient Boosting Regression Trees and Digital Twin platforms, which prolong PV module lifespan (Blackburn et al. 2020). AI-based cooling was shown to achieve up to 6.7% yearly energy savings in industrial applications (Zaman et al. 2023). Moreover, AI maximizes radiative and liquid-based cooling through dynamic adaptation to solar radiation, ambient temperature, and wind speed. AI optimization methods such as deep learning, genetic algorithms (GA), and hybrid models also bring state-of-the-art PV cooling performance. For example, genetic algorithms are used to optimize the flow channel design in PV/T systems, improving both thermal and electrical efficiency. Overfitting risks are mitigated through robust validation techniques such as cross-validation and regularization, ensuring that AI models generalize well across different PV systems and climatic conditions. Furthermore, AI models are increasingly optimized for real-time application, using lightweight architectures or deploying models at the edge, reducing latency and ensuring faster, real-time control in dynamic environments. AI models employed for PV cooling, their optimization goals, and outcomes are summarized in Table 8. These techniques improve efficiency, operational management, and energy conservation, and hence AI is a game-changing technology for sustainable PV energy management.

However, it is important to note that many of these studies primarily report average performance metrics without providing measures of variability such as standard deviation, confidence intervals, or error bars. This omission limits the assessment of performance robustness and reproducibility across different environmental conditions and system configurations. Future research should include such statistical measures to enable a more comprehensive evaluation of AI-based PV cooling performance. AI-powered cooling solutions are transforming photovoltaic (PV) technologies' performance with machine learning, deep learning, and optimization-based techniques. Using real-time control and forecasting capability, artificial intelligence enhances cooling efficacy, saves energy, and increases the life cycle of modules longer than conventional practices. The latest systems adapt according to fluctuating environmental factors, thus ensuring utmost thermal management and operational efficiency. The future technologies can focus on self-learning systems, advanced digital twins, and multi-agent optimization, advancing automation, sustainability, and cost reduction even more. With the application of AI, the renewable energy sector can achieve higher reliability, energy conservation, and long-term PV performance.

4.2 Machine learning models for photovoltaic (PV) cooling

Machine learning (ML) has become a fundamental tool for the enhancement of photovoltaic (PV) cooling systems through its potential to provide accurate predictions, real-time optimization, and energy efficiency improvement. The models improve the performance of PV systems through adaptively controlling cooling parameters, optimizing heat dissipation mechanisms, and reducing energy losses. Machine learning models were employed to forecast and optimize PV/T systems' electrical and thermal performance. For instance, Multi-layer Perceptron (MLP) and Light Gradient-Boosting Machine (Light-GBM) have both obtained outstanding prediction performance for PV module temperature estimation and energy efficiency with R^2 of 0.9906 and 0.983, respectively (Safae et al. 2024). Likewise, Convolutional Neural Networks (CNN) coupled with Firefly Optimization Algorithm have been used to optimize thermosiphon heat pipe operating conditions for enhanced cooling performance (Abdulrahman 2024). Also, Adaptive Neuro-Fuzzy Inference System (ANFIS) models coupled with optimization algorithms like the Manta Ray Foraging Algorithm (MRFOA) have been used to simulate and optimize the performance of PV/T collectors with an improvement in electrical efficiency by 27.7% (Zayed et al. 2024).

Accurate performance modeling also requires integration of thermal effects on the electrical characteristics of PV panels. Elevated temperatures impact the current–voltage (I-V) curves by decreasing the open-circuit voltage (Voc) and increasing the short-circuit current (Isc), generally reducing overall efficiency by about 0.5% per Kelvin increase in temperature (Zaini et al. 2015; Libra et al. 2017; Libra and Poulek 2022). Advanced electro-thermal coupled models dynamically simulate the junction temperature and translate thermal behavior into electrical performance metrics, enabling machine learning models to predict I-V curve shifts more accurately under varying environmental conditions (Dusabe et al. 2008; Mai et al. 2017; Abdulrazzaq et al. 2021). These models support the optimization of maximum power point tracking (MPPT) algorithms—such as Perturb and Observe (P&O), Incremental Conductance (IC), and adaptive

Table 8 AI-based optimization strategies for photovoltaic cooling system

Ref	AI model used	Cooling system	Optimization goal	Results
Shaban et al. 2024)	Optimized Deep Neural Network (ODNN) with Sand Cat Swarm Optimization (SCSO)	PV/T system (Water, Air, Electrical)	Maximize prediction accuracy and operational efficiency	MAE: 0.0075, MSE: 0.00251, R^2: 0.9972; effective in dynamically adjusting hyperparameters
Essa et al. 2023)	Radial Base Function Neural Network with Gorilla Troops Optimizer (RBFNN-GTO)	PV cooling with rectangular cascade channels	Predict key performance parameters (thermal, electrical efficiency)	Decrease in cells' average temperature by 27 °C; improved electrical, thermal, and overall efficiencies by 4.8%, 59.5%, and 73.33% respectively
May Tzuc et al. 2018)	Artificial Neural Networks, Support Vector Machine (SVM), Regression Tree Ensembles	Smart PV module cooling systems	Estimate operating temperature of PV modules (T_{pv})	Precision of about 93%; SVM most appropriate for reflecting physical interactions
Abd Elaziz et al. 2021)	Random Vector Functional Link (RVFL) with Mayfly Optimization (MO)	PV/T system with Electrolytic Hydrogen Production (EHP)	Maximize prediction accuracy for power output, surface cell temperature, etc	RMSE for training: 0.25 (RVFL), 0.65 (RVFL-MO); RMSE for testing: 1.63 (RVFL), 2.04 (RVFL-MO)
Wang et al. 2023b)	Non-dominated Sorting Genetic Algorithm-II (NSGA-II)	PV/T-CCHP system	Optimize equipment sizing and operating strategy	Annual total cost savings: 5.82%; primary energy savings: 22.63%; carbon emissions reduction: 37.51%
Xie et al. 2023)	Multi-objective and multi-stage optimization with penalty factors	Topology-optimized internal flow channel	Improve cooling performance and reduce cell temperature	Solar cell temperature reduction: 13.85%–23.45%; net PV system output power increase: 3.00%–19.37%

fuzzy logic controllers—that maintain optimal power extraction despite temperature fluctuations and irradiance changes (Lingamaiah et al. 2024; Kwan and Wu 2016; Chen et al. 2014). The integration of thermal and electrical modeling frameworks into AI-driven PV cooling systems enhances real-time cooling control and maximizes energy harvesting efficiency. Simulation tools like MATLAB/Simulink are commonly used for implementing and validating these coupled models, showing strong alignment with experimental data (Partho et al. 2023).

Moreover, autoregressive models such as the Auto-Regressive Integrated Moving Average (ARIMA) have been employed to forecast temperature changes of phase change materials (PCM), hence facilitating seamless integration with photovoltaic (PV) cooling systems (Kiaghadi et al. 2024). Reinforcement learning-based strategies, in combination with internet of things (IoT) sensor networks, have also demonstrated potential in real-time adaptive cooling, with response time enhancement by 30% (Williams et al. 2023). The main machine learning methods applied in PV cooling, their optimization objectives, and the respective performance measures are explained in Table 9 below.

It is important to note that many machine learning studies primarily report average performance metrics, such as R^2 and RMSE, without accompanying statistical measures of variability such as standard deviation or confidence intervals. These uncertainty quantifications are critical for evaluating model robustness, preventing overfitting, and ensuring generalizability across varied datasets and operational scenarios. Incorporating such statistical measures in future evaluations will enhance confidence in the reliability and practical deployment of ML-based PV cooling models. Machine learning has transformed PV cooling through smart, data-oriented optimizations for improved efficiency, flexibility, and sustainability. Neural networks ensemble learning, and autoregressive algorithms accurately forecast temperature changes, optimize energy generation, and regulate cooling in real time. These AI-based solutions optimize PV performance while reducing cost and energy consumption. With continued advancements in ML, the incorporation of

Table 9 ML models for PV cooling

ML Model	Application	Performance Metrics	Reference
MLP	Predict electrical and thermal efficiency of PV/T systems	$R^2 = 0.9906$ (electrical), $R^2 = 0.983$ (thermal)	Safae et al. 2024; Jakhar et al. 2023)
LightGBM	Predict thermal energy in PV/T systems	$R^2 = 0.988$	Boujoudar et al. 2024)
GBR	Predict electrical power generation	$R^2 = 0.9709$	Safae et al. 2024)
ELM	Optimize thermal and electrical efficiency of PV/T systems	Outperforms ANNs and GP in accuracy and learning speed	Mojumder et al. 2017)
ARIMA	Predict PCM temperature and melting fraction	RMSE = 0.11, MAPE = 0.0003	Kiaghadi et al. 2024)
ANFIS	PV/T systems with water-based nanofluids	R^2: 0.9534, Electrical efficiency enhancement: 27.7%	Cao et al. 2022)
GPR	PV/T systems optimization	R^2: 0.9987 to 1 for multiple objectives	Anisi et al. 2025)
CNN + Firefly	Optimize thermosiphon heat pipe operational conditions	RMSE = 3.95 W	Abdulrahman 2024)
DT and RF	PV power generation prediction, cooling optimization	High accuracy (99.83%)	Dhanalakshmi et al. 2022)
ANFIS-MRFOA	Hybrid solar thermal systems	$R^2 = 0.989$	Zayed et al. 2024)

real-time monitoring, IoT sensors, and sophisticated optimization will further optimize PV cooling measures. This development is critical to driving sustainable, high-efficiency solar energy systems and keeping PV technology a high-performance choice for renewable energy.

4.3 Smart sensors and IoT in photovoltaic (PV) cooling

The integration of internet of things (IoT) technology and smart sensors has enhanced the performance of photovoltaic (PV) cooling systems via real-time monitoring, automatic cooling, and predictive maintenance. IoT-based systems use temperature, humidity, and irradiance sensors for feedback information, which is computed to optimize the operation of coolants such as active air cooling, water cooling, and thermoelectric modules. Such dynamic cooling systems prevent overheating, ensure optimum panel performance, and extend lifespan. Efficiency improvement has been demonstrated in research through a system that recovered 40% of power loss from high temperatures using IoT-controlled water cooling and another system that provided 9.39% power harvest efficiency improvement using real-time adaptive cooling and monitoring (Pojas et al. 2024; Malhan and Ahmed 2022). In addition, IoT-enabled dust monitoring and cleaning systems provide optimization of solar irradiance exposure, minimizing maintenance costs and optimizing long-term energy output (Mukundaswamy et al. 2024).

Field applications of IoT-enabled cooling technologies are becoming popular in large-capacity solar farms. Researchers have pointed out applications like volatile mist cooling systems and Blynk IoT-based air cooling, which dramatically improve energy generation efficiency (Razak et al. 2021; Al-Jebory et al. 2024). Separate commercial-level solar farms with IoT-enabled cooling lack proper documentation, but similar technologies are under trial at laboratory and industrial levels, suggesting strong potential for large-scale application. Moreover, IoT-enabled Maximum Power Point Tracking (MPPT) algorithms with AI-powered optimization additionally enhance PV cooling efficiency by dynamically controlling the cooling operations based on environmental factors (Rawdhan and Ahmed 2024). With more large solar farms, intelligent cooling control through IoT and intelligent sensors will be crucial for optimizing efficiency, lowering cost of operation, and facilitating higher sustainability in the renewable energy sector.

Additionally, passive cooling in photovoltaic (PV) systems occurs through a coupled mechanism involving both radiative and convective heat transfer. Forced convection,

particularly due to wind, plays a critical role in modulating radiative cooling performance. Increased wind speeds enhance convective heat dissipation, leading to a significant reduction in module temperature—often by several degrees Celsius—which in turn improves electrical efficiency (Tomšič et al. 2025; Palovics and Nemeth 2023). However, elevated wind speeds can also disrupt the thermal boundary layer essential for effective radiative heat loss, thereby diminishing the overall radiative cooling effect (Zhuang et al. 2023; Li et al. 2023).

To capture these complex interactions, Computational Fluid Dynamics (CFD) and conjugate heat transfer models have been widely applied. These models incorporate environmental variables such as wind speed and solar irradiance to simulate and optimize the coupled cooling process (Marwaha et al. 2022). Experimental investigations further support these findings, identifying an optimal wind speed range—typically between 1 and 1.5 m/s—that maximizes the combined efficiency of radiative and convective cooling in PV modules (Zhou et al. 2024; Hamdan et al. 2024). Understanding and leveraging the interplay between radiative and convective cooling is therefore crucial for the design of effective hybrid passive cooling solutions, especially in locations subject to variable wind conditions. Integrating this knowledge enhances the thermal management of PV systems, ultimately contributing to improved performance and longevity.

4.4 Challenges and limitations of AI in PV cooling

Artificial intelligence (AI) has transformed photovoltaic (PV) cooling systems through predictive maintenance, real-time temperature control, and efficiency optimization. In spite of its merits, AI application in PV cooling systems encounters some challenges pertaining to computational requirements, integration with systems, heterogeneity of the environment, and economic considerations. These challenges must be addressed for deployable, low-cost, and stable AI-based PV cooling systems.

One of the key challenges is the high data and computational expenses since artificial intelligence models need huge volumes of good-quality data to train and function. Data collection, processing, and storage involved can be computationally expensive, especially for deep learning-based methods that are computationally intensive (Mateo Romero et al. 2022; Kumar et al. 2024b). In addition, it is challenging to incorporate into current PV cooling systems since it entails sophisticated sensors, real-time data acquisition systems, and stable communication networks (Călin et al. 2024; Hamid et al. 2025). AI-based remote monitoring also makes PV systems vulnerable to cyber-attacks, and thus it demands strong encryption and security protocols (Al-Mayyahi et al. 2024). Moreover, environmental conditions like varying irradiance, ambient temperature, and weather conditions influence the validity and reliability of AI forecasts, requiring the formulation of adaptive algorithms with the ability to adjust dynamically to varying conditions (Shaban et al. 2024; Pathak et al. 2022).

Optimization and control complexity is another significant limitation, where AI models have to balance efficiency, responsiveness, and real-time flexibility. AI algorithms have to handle the nonlinear dynamics of PV cooling processes, and optimal energy consumption and cooling performance is hard to realize (Mateo Romero et al. 2022; Kumar et al. 2024b). Thermal cycling effects, PCM phase-separation, and nanoparticle agglomeration further complicate the long-term performance of cooling systems. Thermal cycling can lead to material fatigue and degradation in PCM thermal conductivity. To address this, multi-layer PCMs with graded nanoparticle distributions have been proposed, offering better thermal buffering and electrical efficiency (Al Arni et al. 2024). PCM phase-separation is mitigated using nano-enhanced PCMs (NePCMs) that improve phase stability and thermal response (Said et al. 2024). Surfactants such as sodium dodecylbenzene sulfonate (SDBS) are also incorporated to prevent nanoparticle agglomeration in PCM matrices, enhancing specific heat, latent heat, and thermal conductivity (Zaimi et al. 2022). Additional enhancements include using fins and cascading PCMs (Awad et al. 2022b), and hybrid nanoparticle blends to balance conductivity and heat storage (Jamil et al. 2023). These combined strategies ensure more robust, longer-lasting cooling performance in AI-integrated PV systems.

Another important consideration is the dynamic energy payback time (DEPT) of active and hybrid AI-based cooling systems. Although these systems enhance electrical and thermal output, they consume energy for operation (e.g., pumps, fans, control units). If not properly designed, they may result in marginal or even negative net energy gains over their operational lifespan. Several studies have evaluated DEPT under different system types and climatic conditions. For instance, general photovoltaic systems show energy payback times (EPBT) ranging between 2 and 4 years, indicating strong net-positive energy gains over time (Black 2005). Hybrid PV/T systems, which simultaneously produce thermal and electrical energy, report EPBT values between 2.9 and 4.7 years (Kumar and Tiwari 2009). High-concentration PV systems such as FLATCON® demonstrate EPBT as low as 8–10 months in high-radiation zones and up to 16 months in less sunny areas like Germany (Peharz and Dimroth 2005). Additionally, dynamic aspects such as declining manufacturing energy input and increased recycling have improved lifecycle EPBT (Zanoni and Mazzoldi 2013). Economic feasibility studies show that depending on local subsidies and electricity prices, hybrid system payback periods can range from 5.3 to 16 years (Liu et al. 2015; Al Hasibi and Haris

2023). These insights support the long-term sustainability and net energy gains of AI-optimized PV systems. However, further techno-economic analysis and life-cycle assessments are recommended to validate their performance across scales and geographies.

While AI-based and PCM-enhanced cooling methods improve PV performance, their sustainability must be carefully assessed through life cycle assessments (LCA). Studies show that integrating PCMs in PV systems increases energy payback times to approximately 14 years, compared to 5 years for conventional PV systems, due to higher initial energy investments and material impacts (Foteinis et al. 2023). The environmental footprint increases by about 21.9% yet gains from improved electrical efficiency and fossil-fuel substitution can provide net environmental benefits over the system's lifetime (Foteinis et al. 2023; Colarossi et al. 2022). Economic viability depends on factors such as electricity pricing and optimal sizing of thermal energy storage units (Ren and Ma 2019). Although direct LCA data on AI-controlled hardware in PV cooling is limited, AI-driven multi-criteria decision-making (MCDM) methods have been proposed to optimize coolant selection and system operation, potentially improving overall sustainability (Chakraborty et al. 2024). These insights underscore the importance of integrating LCA and techno-economic analyses into future work to substantiate sustainability claims for AI- and PCM-based PV cooling solutions.

Economic constraints also hinder the widespread use of AI because high upfront expenses in hardware, software, and expertise make it prohibitive for small- and medium-scale PV installations (Ali et al. 2024b). In addition, scalability is also an issue, with AI models developed for a particular PV system needing significant redesign to be implemented in different climatic zones, types of panels, and cooling systems (Guo et al. 2024).

Solving these challenges demands further development of AI-based thermal control, cost-effective computational models, and more robust machine learning algorithms for enhancing the accuracy and efficiency of predictions. Future research must give importance to hybrid AI models, edge computing to achieve lower latency, and adaptive algorithms that can dynamically adjust based on various PV cooling requirements. By addressing these limitations, AI can be enabled to fully exploit its potential in maximizing PV cooling efficiency, minimizing operational costs, and ensuring long-term solar energy system sustainability.

5 Key challenges and future opportunities in photovoltaic cooling research

As the technology of photovoltaic developed, temperature regulation within PV modules has increasingly become the highest priority when it comes to achieving optimum efficiency and lifespan. Despite many improvements in technology, PV module cooling systems remain significant technical, economic, and environmental issues. However, emerging trends of AI integration, eco-friendly materials, and hybrid system designs offer promising ways to mitigate these challenges and achieve higher system efficiency. Figure 12 encapsulates these challenges and provides possible future research directions for PV cooling.

5.1 Challenges in photovoltaic cooling

5.1.1 Technical challenges

Photovoltaic (PV) systems face significant technical issues, such as thermal degradation, intricate data processing requirements, and the environmental effect of conventional cooling techniques. PV modules experience overheating at high ambient temperature, low wind speed, and high solar irradiance, reaching temperatures of 75 °C. The result is material degradation and module efficiency loss by about 0.45–0.55% for each °C increase in temperature, particularly in hot climates (Kirpichnikova and Makhsumov 2020; Kurtz et al. 2009). Traditional cooling systems, like vapor compression systems, use electricity and refrigerants, which raise the carbon footprint (Li and Strezov 2015; Kumar and Kapilan 2021). While solar-driven cooling systems offer environmentally friendly solutions, their utilization is not widespread.

Moreover, contemporary PV systems also demand advanced monitoring and control systems to manage huge data from sensors and data loggers (Klise and Stein 2017; Wang et al. 2011). Incorporation of AI and machine learning models adds complexities with issues like unavailability of data, overfitting, and intense computational requirements (Klise and Stein 2017). Further, although nanofluids have shown promising heat transfer characteristics, applications are deterred owing to nanoparticle dispersion problems, high cost of production, and environmental issues (Kumar et al. 2022; Thakare et al. 2024). Solving these complex technical issues is imperative to improve the efficiency, sustainability, and scalability of PV cooling technologies.

5.1.2 Economic barriers

The application of sophisticated PV cooling systems is usually deterred by their high economic barriers, beginning with the expensive initial installation costs. Such systems have high up-front costs because the manufacturing and installation of specialized devices like sophisticated heat exchangers and nanofluid-cooling systems are capital-intensive activities (Needleman et al. 2017; Lo et al. 2018). Also, maintenance cost is a significant economic load since such kinds of

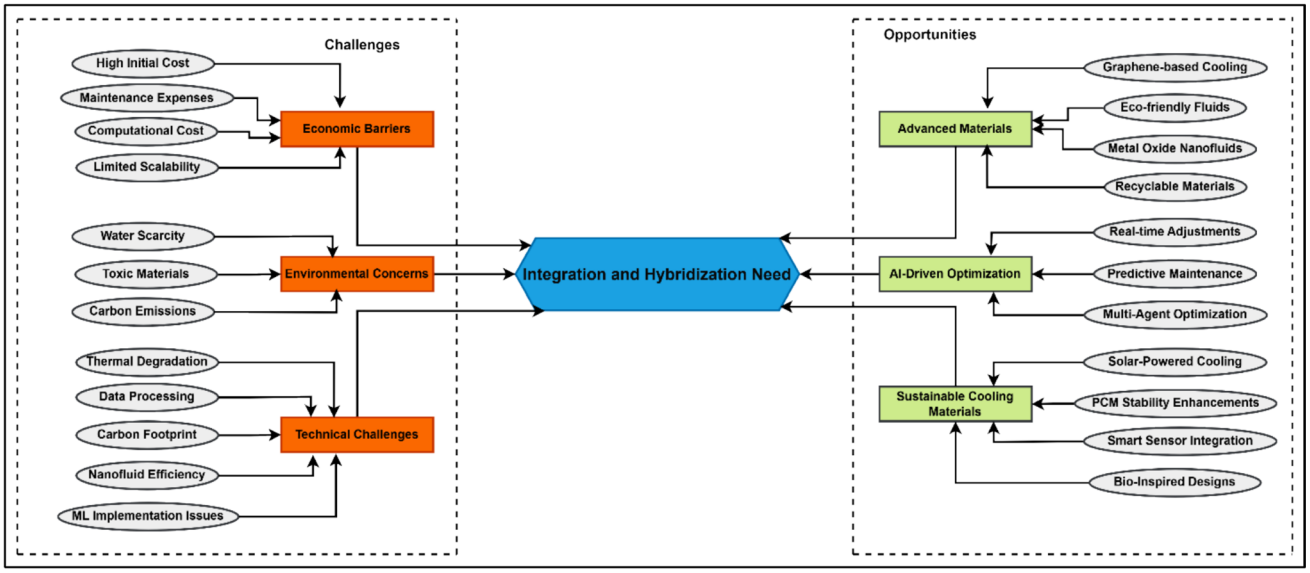


Fig. 12 Key challenges and future opportunities in photovoltaic cooling research

systems generally involve sensitive and sophisticated technologies like PCMs and nanofluid circuits which need special care and continuous maintenance (Ali et al. 2024b; Gao et al. 2023).

Computational cost of AI-driven control systems also adds to economic challenges because real-time optimization and predictive maintenance require high computing hardware (Ali et al. 2024b; Han and Guo 2014). Finally, low scalability owing to infrastructure constraints and the presence of inexpensive, special materials results in scale-up challenges for mass deployment, especially in regions of severe climates or poorly developed supply chains (Hwang et al. 2023; Duck et al. 2018). Overcoming these barriers through the utilization of cost-saving strategies, improved materials, and scalable designs is critical to spur the adoption of PV cooling technologies.

5.1.3 Environmental concerns

Environmental considerations play a critical role in the use of PV cooling systems, especially in areas susceptible to water shortages. Although the water-cooled cooling system is very effective in lowering the temperature of PV modules and improving efficiency, it consumes a huge quantity of water and is, therefore, not feasible in arid and drought-prone regions (Nižetić et al. 2018; Ibrahim et al. 2024). Thus, reducing water use through system optimization is greatly essential to maintain environmental sustainability (Rai et al. 2025). In addition, many nanofluid-based cooling systems incorporate nanoparticles that may be hazardous to the environment and human health if not properly managed (Fthenakis 2018; Tyagi and Kumar 2021).

Moreover, PV cooling systems that rely on non-renewable energy sources for processes such as grid-connected chillers or vapor compression cooling—are contributors to greenhouse gas emissions, cancelling out the environmental benefits of solar energy (Siah Chehreh 2021; Tawalbeh et al. 2021). As countermeasures, employing renewable energy-driven cooling technologies and hybrid cooling systems have been effective solutions to minimize carbon emissions while enhancing system efficiency (Kandeal et al. 2021). Resolving these environmental issues is critical to incorporating PV cooling technologies into a sustainable and low-carbon energy future.

5.2 The need for integrated and hybrid cooling approaches

To overcome technical, economic, and environmental challenges, photovoltaic (PV) cooling research is increasingly moving toward hybrid solutions that combine advanced control systems, eco-friendly materials, and multi-dimensional optimization strategies. Integrating AI and IoT technologies into PV cooling systems has proven effective in enhancing control mechanisms, enabling real-time monitoring, and optimizing system responses to dynamic environmental conditions (Edaris et al. 2024). AI-powered algorithms can adaptively manage cooling operations, improving system efficiency and reducing unnecessary energy consumption. Alongside AI, sustainable materials like eco-friendly nanofluids (e.g., SiO_2-water, Fe_3O_4-water) and phase change materials (PCMs) offer promising pathways to enhance thermal conductivity while reducing environmental impact (El Kassar et al. 2024; Soltani et al. 2020).

In addition, system-level optimization plays a crucial role in achieving balance across technical, economic, and ecological dimensions. Recent studies show that hybrid systems combining cooling methods (e.g., forced convection with PCMs or thermoelectric modules) can reduce peak temperatures and enhance PV panel efficiency by up to 22% (Chandel et al. 2024; Pang et al. 2015). Furthermore, optimization frameworks, such as multi-objective designs for PV/T-assisted desiccant cooling systems (PV/T-DCS), have shown potential in reducing both energy consumption and carbon emissions (Zhang et al. 2025). These integrated hybrid strategies bridge the gap between existing limitations and future innovations, promoting PV cooling solutions that are not only technically viable but also environmentally sustainable and cost-effective.

5.3 Future opportunities for PV cooling

5.3.1 Advanced materials

The development of advanced materials plays a crucial role in enhancing the efficiency and sustainability of PV cooling systems. Among these, graphene-based cooling has attracted significant attention due to graphene's exceptional thermal conductivity, which is superior to conventional metals like copper (Hu et al. 2024; Pavía et al. 2021). Graphene nanofluids and graphene-copper composites have shown remarkable potential in improving heat dissipation in PV modules. However, challenges such as high viscosity and production costs still limit their widespread application (Pavía et al. 2021). Similarly, the transition to eco-friendly fluids is vital as water-based and biodegradable nano-fluids are being explored to replace harmful nanomaterials, contributing to both thermal efficiency and environmental sustainability (Mariam et al. 2024; Shahverdian et al. 2021).

In addition, metal oxide nanofluids such as Al_2O_3-water, TiO_2-water, and ZnO-water have demonstrated superior heat transfer properties, resulting in electrical efficiency improvements of up to 17% in PV cooling applications (Shahverdian et al. 2021; Ahmed et al. 2022). Their thermal performance can be optimized by tuning nanoparticle concentration, shape, and size (Yogaraj et al. 2023). Finally, integrating recyclable materials such as graphene-coated components offer dual benefits of enhancing system efficiency while reducing environmental impacts through material reuse (Bo et al. 2020). These advanced materials represent a significant opportunity to bridge the gap between thermal performance and sustainable PV cooling solutions.

5.3.2 AI-driven optimization

AI-driven optimization plays a transformative role in advancing PV cooling systems by enabling real-time adjustments, predictive maintenance, and multi-agent energy management. Machine learning models, such as Multi-layer Perceptron (MLP) and Light-GBM, are increasingly being used to predict thermal and electrical performance of PV/T systems, allowing for real-time control and dynamic energy matching between PV generation and building loads (Safae et al. 2024; Li et al. 2021). These models improve system responsiveness by continuously adjusting cooling mechanisms in response to environmental variability. Additionally, predictive maintenance powered by AI and IoT platforms helps detect faults early by analyzing trends in PV system data, such as temperature and voltage anomalies (Marangis et al. 2024; Kayalvizhi et al. 2024). Algorithms like e-Xtreme Gradient Boosting and Prophet have proven effective in forecasting system failures, improving up time and reducing operational costs.

Further, multi-agent optimization systems (MAS) are increasingly employed to balance cooling efficiency and energy consumption across distributed PV networks and microgrids. MAS frameworks coordinate multiple components—such as cooling units, energy storage, and PV generation—through hierarchical and decentralized decision-making, improving overall system efficiency and flexibility (Hao et al. 2014; Xiong et al. 2020). This approach ensures that PV cooling systems adapt to varying loads and environmental conditions while minimizing energy wastage. By integrating AI-driven solutions, future PV cooling systems can achieve optimized energy use, lower maintenance costs, and enhanced system longevity—paving the way for more resilient and sustainable solar power infrastructures.

5.3.3 Sustainable cooling methods

Sustainable cooling solutions are gaining momentum in PV systems to reduce grid dependency and environmental impact. Solar-powered cooling systems, such as photovoltaic-thermal (PVT) and PV-driven air conditioners, can harness renewable energy for thermal management, improving PV efficiency and reducing reliance on external power sources (Ali et al. 2024b; Li and Wang 2016). Complementing this, advancements in phase change materials (PCMs)—such as nanoparticle-enhanced PCMs and metal foam composites—have improved thermal conductivity and energy storage capacity, addressing traditional issues like phase segregation and low heat transfer rates (Nandi and Biswas 2025; Reddy et al. 2018).

Moreover, smart sensor integration through IoT platforms enables real-time monitoring and automated control of cooling systems, optimizing energy usage and enhancing operational performance (Figueiredo et al. 2021; Jagannathan et al. 2024). Additionally, bio-inspired designs, such as passive radiative cooling materials that mimic natural heat dissipation processes, offer energy-efficient, power-free

thermal management solutions for PV panels (Jaramillo-Fernandez et al. 2024; Hu et al. 2022). Collectively, these sustainable methods not only improve cooling efficiency but also contribute to the long-term resilience and sustainability of PV systems.

6 Conclusion

Photovoltaic (PV) cooling has emerged as a critical field in enhancing the performance, efficiency, and lifespan of solar energy systems, especially as global demand for renewable energy grows. This comprehensive review highlights the diverse array of cooling techniques, including passive, active, hybrid, and AI-driven strategies, each contributing uniquely to mitigating heat-induced efficiency losses in PV modules.

Passive techniques, such as radiative cooling, phase change materials (PCMs), and natural convection, offer energy-free thermal regulation with moderate efficiency gains but are often limited by environmental conditions and material properties. Active cooling methods, including water- and air-based systems, provide more significant temperature reductions and performance improvements but are constrained by resource consumption and operational complexity.

The integration of passive and active systems through hybrid approaches addresses many of these limitations, with combinations such as PCM-nanofluid and TEC-PCM systems offering enhanced heat dissipation and stable thermal management. Furthermore, the application of AI and machine learning algorithms has revolutionized PV cooling optimization, enabling real-time, adaptive control that balances efficiency with resource use. These intelligent systems are capable of dynamically responding to environmental fluctuations, thereby maximizing PV performance and reducing operational costs.

Despite the progress, several challenges persist. Economic feasibility, environmental sustainability, material degradation, and system scalability continue to hinder the widespread adoption of advanced cooling strategies. Additionally, AI-driven solutions, while promising, require further development to address computational demands, cybersecurity risks, and complex integration.

Looking ahead, future research should prioritize the development of cost-effective, scalable, and eco-friendly cooling technologies. Particular attention should be given to hybrid and AI-integrated systems that offer both high performance and adaptability to diverse climatic conditions. Advancements in sustainable nanomaterials, smart control algorithms, and multi-objective optimization will be pivotal in shaping next-generation PV cooling solutions that support the global transition to clean and efficient solar energy.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declared that no known competing interest is associated with this study from any organization or collaboration that could affect this process.

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